

Can Aid Buy Foreign Public Support? Evidence from Chinese Development Finance

LUKAS WELLNER

Monash University and Georg-August University Göttingen

AXEL DREHER

Heidelberg University, KOF Swiss Economic Institute, Centre for Economic Policy Research, Center for European Studies, and CESifo

ANDREAS FUCHS

Georg-August University Göttingen and Kiel Institute for the World Economy

BRADLEY C. PARKS

William & Mary and Center for Global Development

AUSTIN STRANGE

University of Hong Kong

We have received helpful comments from Antoine Boucher, Mitchell Downey, Christopher Kilby, Friederike Lenel, Cleo O'Brien-Udry, Michael Tierney, and participants at the following conferences, workshops, and seminars: the Beyond Basic Questions Workshop at the Kiel Institute for the World Economy (June 2019), the Globalization and Development Workshop at the University of Göttingen (October 2020), the Conference of the International Political Economy Society hosted by Ohio State University (October 2020), the Graduate Student International Political Economy Virtual Conference "The International Political Economy of Aid and Development" (December 2020), the Meeting of the European Public Choice Society hosted by the University of Lille (April 2021), the 3rd International Conference on Globalization and Development hosted by the University of Göttingen (July 2021), the German Development Economics Conference at the University of Hohenheim (June 2022), the Political Economy of International Organization Conference at the University of Oxford (July 2022), the Centre for Economic Policy Research International Lending and Sovereign Debt RPN "China in the International Financial System" Workshop hosted by the Kiel Institute for the World Economy (June 2023), and the 15th Chinese Economic Association (CEA; Europe) and 34th CEA (UK) Annual Conference "Sustainable Development in a Changing Era: China and the World" (July 2023). We thank Conrad Ning, Xianglu Peng, Yuchen Tang, Hanxiong Yan, Yiling Zhang, Aozuo Zheng, Jing Zhou, and Xinman Zou for excellent research assistance. Axel Dreher and Andreas Fuchs are grateful for financial support from the German Research

Electronically published January 8, 2025

Economic Development and Cultural Change, volume 73, number 2, January 2025.

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<https://doi.org/10.1086/729539>

I. Introduction

A large literature examines the effects of foreign aid.¹ Most of this research focuses on whether or not aid affects tangible outcomes in recipient countries, such as economic growth, health, education, and corruption.² These outcomes are certainly important to those living in developing countries and to development-finance institutions. However, recipient country welfare is only one objective that motivates bilateral donors to provide foreign aid (Thiele, Nunnenkamp, and Dreher 2007).³ Donor countries also use bilateral aid to pursue their own geostrategic goals, such as buying votes in international organizations, supporting friendly governments before elections, securing market access for exporters, deterring asylum seekers, and fighting terrorism (e.g., Kuziemko and Werker 2006; Fleck and Kilby 2010; Faye and Niehaus 2012; Dippel 2015; Rommel and Schaudt 2020). Another key motivation for bilateral aid is the acquisition of soft power—for example, to influence international public opinion about the donor government.⁴

To this end, donor governments spend a considerable amount of time and money disseminating positive messages about their generosity to members of the public in developing countries. They attach their logos to aid shipments. They place signage at project sites to inform the public of their activities. They organize public ceremonies to mark the start of new projects and the completion of existing ones. Some broadcast their own messages through social media

Foundation (DFG) in the framework of the project “Empirical Analyses of Emerging Donors in Development Cooperation” (DR 640/5-3 and FU 997/1-3). Lukas Wellner acknowledges generous support from the DFG in the framework of the Research Training Group 1723 “Globalization and Development” (no. 17867/1222). Bradley Parks acknowledges generous support from the Ford Foundation (no. 134306), the Smith Richardson Foundation (no. 2021–2516), and the William and Flora Hewlett Foundation (no. 2020–1573). We also thank Harvard University for generously providing access to data from the Gallup World Poll. Contact the corresponding author, Andreas Fuchs, at mail@andreas-fuchs.net.

¹ For ease of exposition, we will use the term “aid” in this paper to refer broadly to any types of official-sector financial flows from a donor (or lender) to a recipient (or borrower). In cases when we wish to reference the narrower (Organization for Economic Co-operation and Development’s Development Assistance Committee) definition of aid, we use the term official development assistance (ODA). In cases when we wish to reference concessional and nonconcessional official financing that does not qualify as ODA, we use the term other official flows (OOF).

² See Werker, Ahmed, and Cohen (2009), Dreher, Lang, and Ziaja (2018), and Doucouliagos (2019) for literature surveys.

³ Also, this objective is probably better addressed via multilateral institutions (Milner and Tingley 2010).

⁴ Soft power is “the ability to achieve goals through attraction rather than coercion” (Nye 2004, x). Public opinion is a commonly used proxy for soft power (e.g., Nye 2004; Goldsmith and Horiuchi 2012; Rose 2016).

channels and cultivate journalists to encourage media coverage of their accomplishments. Others are more aggressive, forging content-sharing partnerships with radio stations, television channels, and newspapers or building telecommunication systems that make it easier to transmit information to the general public. In short, aid is used to shape perceptions on the ground in developing countries, and “brand management” is one of the most important reasons why donor governments extend foreign aid bilaterally rather than multilaterally.⁵

Economists and political scientists have estimated the effects of development finance on vote buying, migrant deterrence, the fight against terror, and public opinion in recipient provinces or countries (e.g., Bandyopadhyay, Sandler, and Younas 2014; Lanati and Thiele 2018; Dreher et al. 2019b; Eichenauer, Fuchs, and Brückner 2021). Yet, no study has comprehensively tested whether and to what extent development finance affects overall levels of approval for donor governments. This is a surprising omission because soft power is an important first-order outcome for a number of other strategic goals. Goldsmith and Horiuchi (2012) suggest that foreign public opinion affects military support by foreign countries. Guiso, Sapienza, and Zingales (2009) and Rose (2016, 2019) show that soft power and higher levels of trust between countries also bring material economic gains, such as higher exports for countries with greater global influence. Disdier and Mayer (2007) find stronger trade ties between countries whose populations have higher levels of affinity for each other. What is more, to the extent that soft power affects economic outcomes in the donor country (via increased trade, for example), it may increase support for aid giving in the donor country and as such also lead to higher volumes of future foreign aid flows to recipient countries.

Whether and how aid improves foreign public perceptions of governments is of growing importance for many of the largest bilateral donors. As Goldsmith, Horiuchi, and Wood (2014, 88) point out, “Competition between major powers such as the United States (U.S.) and China for favorable perceptions in global public opinion is increasingly evident today and likely to be a pivotal feature of the emerging international order.” However, whether aid expands or erodes support for donor governments abroad remains an open question.

Instead of bolstering support for donor governments, development projects could easily become reputational liabilities if they are not carefully designed and implemented. Projects that involve large-scale construction activities often create noise, traffic, and pollution. They can lead to labor strikes, public

⁵ They do so in spite of well-documented concerns related to aid proliferation and fragmentation (Knack and Rahman 2007; Gehring et al. 2017).

protests, lawsuits, and allegations of political favoritism and corruption.⁶ Additionally, development projects can backfire—from a “brand-management” perspective—if they fail to reach completion or experience major cost overruns that are borne by local and national governments.⁷

Potential public opinion effects are not limited to the localities where aid projects occur. Aid branding and publicity can affect attitudes in farther-flung places too, and attitudinal effects in some places might offset effects in others. Gauging the effects of foreign aid projects on a donor’s soft power therefore requires an estimate of aggregate effects in addition to partial, localized estimates that are specific to project sites.

We investigate the effects of development projects on a donor government’s popular support across three different target audiences: (i) people living in the province(s) where a project takes place, (ii) people living in other areas of the country with less direct exposure to the project, and (iii) international audiences without direct exposure to projects. These three levels of analysis allow us to distinguish between direct and indirect public opinion effects. Provinces that host development projects have the highest levels of “treatment exposure,” and the people living within these jurisdictions are most directly affected by positive and negative project outcomes. Those who live in close proximity to development projects will be more likely to make judgments about the donor government on the basis of their own firsthand experiences and observations—or those of the people whom they know. However, indirect public opinion effects can also occur when people inside or outside the province where the project is located learn about it via television, radio, print media, online media, word of

⁶ For example, during the middle of winter in 2018, a China Eximbank–financed thermal power plant in the Kyrgyz Republic failed, and local residents were left with no heating. When civil society organizations followed the paper trail, evidence of embezzlement emerged. This resulted in the dismissal of Prime Minister Sapar Isakov. Thirty government officials were charged with corruption and using their positions to lobby for the selection of a Chinese company (TBEA) as the contractor for the project. Prosecutors estimate that bid rigging and the inflated cost of the sole-source contract issued to TBEA cost the government of the Kyrgyz Republic as much as US\$111 million (Malik et al. 2021).

⁷ The Astana Light Rail Construction Project is a case in point. China Development Bank issued a US\$1.5 billion loan to Astana LRT LLP—a project company that is owned by the City of Astana—for this project, and the government of Kazakhstan provided a sovereign guarantee in support of the loan. However, in October 2019, the president of Kazakhstan ordered an investigation into the officials who initiated the project. The chief executive of Astana LRT LLP was accused of embezzling project funds and fled the country. The local authorities suspended the construction of the railway, and the half-finished project became a source of public discontent (Malik et al. 2021). The 4-meter-high concrete trestles upon which the railway was supposed to run are now referred to by local residents of Astana as “monuments of corruption.”

mouth, or travel. Such effects are, of course, not restricted to residents of the host province or even country, and different audiences around the world can react differently to information about development projects. The construction of Hambantota Port in Sri Lanka is a case in point. This project, which was financed by China Eximbank, has been cited in thousands of media reports in virtually every corner of the globe as evidence that the Chinese government is engaging in “debt-trap diplomacy” (e.g., Brautigam 2020).⁸ Nor do development projects necessarily produce consistent public opinion effects across these different audiences. For example, a development project that is considered to be useful and appropriate by residents of the recipient country might elicit a very different response to the donor or recipient from observers in a rival country, who may view the donor’s or recipient’s gains as coming at their expense.

We focus our empirical analysis on Chinese-government-financed development projects. Beijing’s overseas development program is a useful application for several reasons: China has become the world’s largest bilateral source of international development finance. It now outspends the United States on a more than 2-to-1 basis.⁹ Like other major powers, China is increasingly seeking to expand its economic and political influence around the world. Development finance is an important tool that the Chinese government uses to burish its popular image in the Global South (Kurlantzick 2007; Hanauer and Morris 2014a, 2014b; Fuchs and Rudyak 2019). In 2014, Chinese president Xi Jinping acknowledged that the Belt and Road Initiative (BRI) is part of a broader effort to “increase China’s soft power, give a good Chinese narrative, and better communicate China’s message to the world” (*People’s Daily* 2014). Echoing this point, a senior Chinese government official announced that “the work of foreign aid relates to China’s image. We cannot tolerate any negligence

⁸ Another prominent example is the China Eximbank–financed Entebbe Airport Upgrading and Expansion Project in Uganda, which became a major source of international controversy when various media outlets reported (incorrectly) that the airport was a source of collateral the lender could seize in the event of default. Beijing tried to put the issue to rest by issuing the following public statement: “Not a single project in Africa has ever been confiscated by China because of failing to pay Chinese loans.” But Beijing was lampooned by Trevor Noah—the host of a satirical television news program called *The Daily Show*—for the careful wording of its statement. In a video clip that has now been viewed nearly 4 million times on YouTube, Noah said: “I don’t know, maybe it’s just me, but that statement was not the most reassuring thing I’ve ever heard because ‘We’ve never confiscated an airport’ is very different from ‘We’re never going to confiscate an airport’” (Parks, Malik, and Wooley 2022).

⁹ Whereas average annual development finance commitments from China amounted to US\$85.4 billion between 2013 and 2017, average annual development finance commitments from the United States amounted to US\$37 billion during the same 5-year period (Malik et al. 2021; Dreher et al. 2022).

or projects of poor quality” (MOFCOM 2014). The Chinese government is also attractive from an inferential leverage perspective because it is the only bilateral donor for which all development projects have been subnationally geocoded for a substantial number of years across all major world regions. These data—in conjunction with public opinion data that have broad spatio-temporal coverage—provide a strong empirical foundation for the identification of potential causal effects.

We measure the causal effects of Chinese development projects in the short and longer runs. First, we use time-stamped, respondent-level data for more than 1.5 million people interviewed by the Gallup World Poll across 126 countries between 2006 and 2017 in an event study to analyze the short-term effects. This approach exploits the staggered rollout of the poll and the availability of precise interview dates. The precise dates allow us to compare respondents who were interviewed within 30 days before versus 30 days after the occurrence of a Chinese project event. We create a new database coding such events that includes 3,998 commitment, start, and end dates of 2,214 Chinese development projects in 126 countries and 2,025 first-order subnational administrative (ADM1) regions around the world.¹⁰ Controlling for province-year fixed effects, as well as a range of individual and survey characteristics, the timing of an interview can be considered random relative to a Chinese project event. This research design enables us to rigorously analyze the immediate effects of development projects on public attitudes.

While the event study enables estimation of public opinion effects at the discontinuity and thus facilitates the identification of causal effects, it comes at the cost of neglecting a large share of available data. This is because it relies only on projects with event dates that occur during Gallup survey windows. What is more, these estimates relate to the specific timing of project events and are thus short term in nature. This approach does not capture the potential longer-term attitudinal effects of Chinese-government-financed projects.

The second component of our empirical strategy uses an instrumental variables approach to test longer-term effects and to more comprehensively assess the public opinion effects of China’s overseas development projects. We again make use of georeferenced and temporally disaggregated Chinese-development-project data and subnational public opinion data from the Gallup World Poll. We estimate the effects of Chinese development projects on public approval of

¹⁰ ADM1 regions are one layer below the national level and correspond, for example, to provinces, states, oblasts, governorates, and emirates, depending on the administrative divisions in place in a given country.

Beijing not only within recipient provinces and countries but also across the Global South as a whole.

Our instrumental variables strategy follows Dreher et al. (2021b) and Bluhm et al. (2025) and makes use of a supply shock—the yearly production volumes of physical construction materials produced in China—to proxy the over-time availability of Chinese projects. China overproduces materials, such as steel, relative to its domestic demand (Dreher et al. 2021a). Chinese-government-financed development projects are often tied to goods and services provided by Chinese companies, and, as such, they also heavily rely on input materials produced in China. Therefore, larger production volumes of construction materials in China should increase the supply of overseas development projects. Dreher et al. (2021b) and Bluhm et al. (2025) use the share of years over the sample period in which a region received a development project from China to proxy which regions are likely to receive larger or smaller shares of additional projects that result from these supply shocks. The instrumental variable is the interaction of the supply-shock measure with this probability of receiving aid.

This identification strategy is based on an intuition similar to that of a difference-in-differences design. We investigate a differential effect of Chinese project input surpluses on public opinion in provinces that have different exposure levels to Chinese development projects. The identifying assumption is that, apart from the direct effect of Chinese development projects on public opinion, public opinion in provinces with differing probabilities of receiving development finance from the Chinese government will not be differentially affected by changes in China's production of physical project inputs, after controlling for province-year fixed effects and country-year fixed effects and the other variables in the model. Below we provide tests of several underlying assumptions needed to make this approach plausible. Our results show that the completion of Chinese development projects increases popular support for the Chinese government in recipient countries. This finding is consistent across the event study that captures effects of exposure to Chinese development projects within narrow, 30-day windows at the discontinuity and in the annual, macro-level analysis that includes a larger sample of projects. On average, we estimate that the completion of one additional Chinese development project increases public approval for the Chinese government at the recipient-country level by more than 3 percentage points in the short run and 0.2 percentage points in the longer run.

We test the mechanisms behind the longer-term findings by looking at the potentially positive and negative side effects of Chinese development projects. The results suggest that Chinese development projects tend to foster development in

recipient countries through higher reported and perceived incomes, improvements in living standards, and higher levels of satisfaction with public goods provision.¹¹

Beyond these country-level effects, a donor country's ability to amass soft power depends on global perceptions of its development projects. Analyzing reactions across all developing countries included in our sample, we find that China's provision of development projects increases public support for the Chinese government among countries in Africa, potential "swing states" in the United Nations General Assembly, and countries that have higher baseline (ex ante) levels of public support for the Chinese government. All of these countries are arguably "high-value" targets for Beijing. We also find that Chinese development finance increases support for the Chinese government in third countries (i.e., neither the donor country nor recipient country) that are politically aligned with the respective recipient country. These indirect gains make development finance an especially attractive instrument for the accumulation of soft power. On the African continent, for example, we estimate that Beijing's global project portfolio increases public approval of the Chinese government by more than 2.2 percentage points per year, on average. Finally, our results show that Chinese development finance boosts public approval ratings for the governing authorities in recipient countries, which, from a political perspective, is consistent with the "win-win cooperation" principle that is commonly used by the Chinese government to describe its foreign aid program.

Our findings contribute to the aid-effectiveness literature in general and the literature on aid and public opinion in particular (e.g., Milner and Tingley 2013; Dietrich, Mahmud, and Winters 2018). They also add to a growing strand of research that measures the intended and unintended effects of Chinese development finance. Recent work has shown that Chinese development projects increase economic growth and reduce the spatial concentration of economic activity, but they also fuel local corruption, stoke ethnic tensions, weaken trade-union participation, instigate public protests, and degrade the natural environment (Isaksson and Kotsadam 2018a, 2018b; Isaksson 2020; Dreher et al. 2021a, 2021b; Iacolla et al. 2021; Gehring, Kaplan, and Wong 2022; Baehr, BenYishay, and Parks 2023; Bluhm et al. 2025; Marchesi, Masi,

¹¹ China's public opinion gains are less rather than more pronounced among people who live in close proximity to completed Chinese development projects. Relative to the country level, more citizens in project provinces report a deterioration of their living standards and a drop in their perceived incomes after the completion of Chinese development projects. This may be the result of lower-than-expected project quality, which is likely experienced to a greater degree among the residents of project provinces.

and Paul, forthcoming).¹² Likewise, recipient governments seem to benefit from these projects because they can steer funds to politically consequential jurisdictions in order to advance their electoral interests (Dreher et al. 2019a; Anaxagorou, Efthymoulou, and Sarantides 2020).

Our work is most directly related to a growing set of studies that investigate the effects of aid on China's image within recipient countries. These studies typically analyze local public opinion effects around project sites—with mixed results. Using geocoded data on Chinese development projects and Afrobarometer survey data, Blair, Marty, and Roessler (2022) leverage a spatial difference-in-differences strategy to estimate the public opinion effects of local exposure to Chinese development finance in 38 African countries. They find that Chinese development projects are disproportionately announced in places that hold positive views of China, but individuals who live near completed projects report less favorable views of China and more favorable views of the United States and other Western powers. In contrast, a recent study finds that the general public in Africa mostly provides positive evaluations of China's foreign aid, particularly within host countries that are more developed and democratic (Han and Huang 2016). Similarly, a separate study using Afrobarometer data decomposes African perceptions toward Chinese trade, investment, and aid and finds that the latter two are often viewed more positively by African citizens (Morgan 2019). Finally, Chen and Han (2022) find that individuals more supportive of incumbent political parties are most supportive of aid from China as well.

Our paper is unique in several ways. We study public opinion gains and losses that accrue locally as well as in other areas of recipient countries and in other countries. As such, we offer a comprehensive assessment of the effects of aid on foreign perceptions of donors. In addition, previous work also focuses almost exclusively on attitudes toward Chinese projects in African countries. One study finds no evidence at national and subnational scales that China's trade, aid, and investment in 18 Latin American countries affect public attitudes toward Beijing, on average (Eichenauer, Fuchs, and Brückner 2021).¹³ But China's development finance is a global phenomenon, and Asia is the largest recipient of Chinese development finance (Dreher et al. 2022). Our study analyzes a substantially larger sample of development projects that covers all developing regions, with an identification strategy that enables estimation of causal effects in regressions at the level of individuals, provinces, and

¹² Dreher et al. (2022) provide an overview of this literature.

¹³ They do, however, find evidence of a polarization effect, with "treated" individuals being more likely than "untreated" individuals to express very positive or very negative views of China—consistent with the view that Chinese development projects create winners and losers.

countries.¹⁴ As such, we can analyze the effect of development finance on soft power at various levels. We use individual-level data to estimate short-term effects of aid events on donor approval. Then, we use province-level and country-level aggregates to examine the longer-term effect of aid on Chinese government approval in project provinces, recipient countries, and at the global level. Relying on the staggered rollout of the Gallup World Poll at the micro level and an instrumental variable for aggregate-level analyses allows us to report causal effects rather than just conditional correlations. In addition, the analysis on the global level also takes into account that China might value popular support in some developing countries more than in others. Ultimately, this nuanced view of the overall level of soft-power gains is needed when it comes to evaluating the effectiveness of aid as a tool to enhance global soft power. This makes our study the first comprehensive analysis of the soft-power effects of foreign aid via public opinion.

We proceed as follows. Section II introduces our new data on Chinese project events and presents the survey data on the approval of China's government. In section III, we analyze short-run effects with the individual-level event study. In section IV, we proceed with the analysis of longer-run effects at the province and country level that relies on the instrumental variables strategy. Section V studies effects at the global level. Section VI analyzes whether Chinese development finance is "win-win" cooperation in the sense that it also raises citizens' approval of the national government of recipient countries. Finally, section VII concludes.

II. Data

A. *New Data on the Timing of Chinese Development Projects*

Analyzing the effects of China's development projects on public approval of the Chinese government requires comprehensive project-level information with high spatial and temporal precision. Such data are unavailable via official channels because the Chinese government does not release project-level information of this nature. The Ministry of Commerce—China's lead institution

¹⁴ Jones (2018), Xu and Zhang (2020), and Blair, Marty, and Roessler (2022) leverage variation in the timing of project implementation, comparing individuals interviewed in locations before the project implementation phase with those interviewed in locations where projects started the implementation phase. To the extent that pledges and commitments create expectations about outcomes that implemented projects do not meet, this approach biases the estimated coefficients downward because it cannot disentangle the effects of announcement and delivery. What is more, the assumption that the timing when projects are committed to relative to when they start being implemented is random relies on the absence of time-varying effects on projects and opinion, which might or might not hold true. Eichenauer, Fuchs, and Brückner (2021) use a plausibly exogenous instrument at the country level but do not report causal estimates at finer scales.

responsible for foreign aid during the past several decades—ranks last among the 47 international donors evaluated in the 2020 Aid Transparency Index (Publish What You Fund 2020). China's State Council publishes official white papers on foreign aid, but the information in these publications is largely limited to aggregate statistics by world regions and decades (State Council 2011, 2014, 2021).¹⁵ In response to this lack of official data, several open-source research initiatives have created project-level datasets that combine and refine information contained across government documents, media reports, and registrars maintained by recipient governments and international organizations (e.g., Strange et al. 2017a; Ray et al. 2021). AidData's Global Chinese Official Finance Dataset, created using the Tracking Underreported Financial Flows (TUFF) methodology, is to date the most comprehensive data-gathering effort because it covers the “known universe” of China's development projects in developing countries (Dreher et al. 2021b, 2022).¹⁶ In addition to dozens of other variables, it provides detailed project descriptions, classifications according to sectors and flow types, and—in 56% of the project cases—information on monetary commitment amounts.

We draw on the 1.1.1 version of this dataset, in which Bluhm et al. (2025) geocoded all implemented and completed projects included in AidData's Global Chinese Official Finance Dataset, in order to estimate localized effects by using precise geographic information on development projects.¹⁷ Bluhm et al. (2025) assigned longitude and latitude to each project site where possible. In sum, the data contain 3,485 Chinese development projects worth US\$273.6 billion (in terms of committed finance) implemented in 6,190 project locations across 134 countries. These locational data allow us to attribute 2,183 projects to the level of ADM1 regions, which typically correspond to provinces or states and are the most fine-grained level of analysis that we require for our study design.¹⁸ A total 1,519 of these projects can be classified as official development assistance

¹⁵ Fuchs et al. (2022) introduce a method to track Chinese aid exports with official customs trade data.

¹⁶ This dataset is a widely used data source in analyses that examine the nature, allocation, and effects of Chinese development finance. Contributions in economics and political science include Hsiang and Sekar (2016), Hernandez (2017), Isaksson and Kotsadam (2018a, 2018b), Anaxagorou, Efthymou, and Sarantides (2020), Isaksson (2020), Martorano, Metzger, and Sanfilippo (2020), Eichenauer, Fuchs, and Brückner (2021), Horn, Reinhart, and Trebesch (2021), Iacolla et al. (2021), Zeitz (2021), Cervellati et al. (2022), Gehring, Kaplan, and Wong (2022), and Watkins (2022), among others.

¹⁷ In September 2021, AidData presented an updated version of this dataset, which also covers the years 2015–2017 (Custer et al. 2021; Dreher et al. 2022). However, the lack of precise geographic information prevented us from using these data at the time we performed our analysis. Such information has meanwhile been provided in Goodman et al. (2024).

¹⁸ We thus use project locations with precision codes 1–4. Precision code 1 corresponds to an exact location; precision code 2 corresponds to locations within 25 km of the exact project site; precision

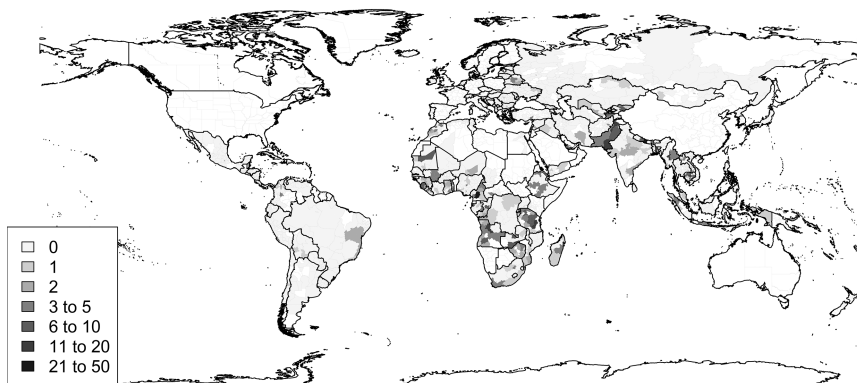


Figure 1. Chinese development projects completed at the province level, 2006–17. Shown is the number of China's development projects by province (ADM1 region) that contain information on their completion date. Sources: Dreher et al. (2022), Bluhm et al. (2025), and authors' data.

(ODA) according to Organization for Economic Co-operation and Development (OECD) definitions.¹⁹ The remaining 664 projects are categorized as other official flows (OOF), meaning they lack concessionality or development intent and are thus more commercially oriented.²⁰ The world map in figure 1 visualizes the number of projects completed over the sample period at the province level.

These data enable researchers to analyze Chinese-government-financed development projects with a level of spatial precision that is not possible for projects financed by any other bilateral donor over such a long period of time.²¹ However, AidData's Geocoded Global Chinese Official Finance Dataset lacks the precise temporal data needed to carefully analyze whether Chinese projects affect public opinion. While all of the project records include information

code 3 corresponds to a second-order administrative (ADM2) region; and precision code 4 corresponds to an ADM1 region.

¹⁹ The dataset codes all Chinese-government-financed projects as ODA if they are financed by Chinese government institutions, have development intent, and have a minimum level of concessionality with a 25% grant element or larger.

²⁰ We include projects that are classified as "vague official finance" in this category. These project records contain insufficient information about concessionality and/or development intent to determine whether they are ODA or OOF.

²¹ At the time we performed our analyses, other global, geocoded project-level databases for bilateral donors we were aware of covered Indian development finance for a period of 8 years (Asmus et al., forthcoming) and French development cooperation for 7 years (AFD 2021). Coverage of the variable capturing attitudes toward these donor governments is with 126,927 and 273,601 observations substantially lower than for the Chinese government (580,484, excluding "don't know" and "refused" answers). For the United States and Japan, geocoded data have also been available for the subset of humanitarian aid but not for overall flows (Bommer, Dreher, and Perez-Alvarez 2022). Bomprezzi et al. (2024) have meanwhile released geocoded data for a large number of Western bilateral donors, which would allow analyses similar to those we provide here (see <https://godad.mc/>).

on the commitment year, information on the exact start and end dates are missing for about three-quarters of the projects, and no calendar-day-level information is available on commitment dates. As such, existing studies have relied on annual-level tests of project allocation and effects. This is a particularly consequential limitation for studies focused on the effects of development projects on outcomes such as public opinion, because such outcomes can be confounded by a host of factors. Causal inference is easier when it is possible to identify narrow windows of time during which other factors should not vary much if at all. What is more, in order to study the effects of completed projects, previous studies had assumed average duration times from commitment to completion (e.g., Dreher et al. 2021b; Bluhm et al. 2025) or only used a small fraction (approximately 25%) of the projects with known start and end dates (Isaksson and Kotsadam 2018a, 2018b; Isaksson 2020). While the former approach assumes no systematic variation in the speed of implementation across different types of projects, the latter assumes that those projects with known start and end dates are representative of the larger sample of Chinese-government-financed development projects. Neither of these assumptions are likely to hold true.

To address this limitation, we revisited the initial data-collection process implemented by AidData (Dreher et al. 2021b, 2022). Our objective was to identify precise commitment, start, and end dates. Commitment dates represent grant and loan agreement signings or official project announcements. Start dates are typically marked by foundation-stone laying and other forms of groundbreaking ceremonies. End dates typically mark the end of construction, the point at which new infrastructure can be used, and/or the date on which a project passed final inspection. The TUFF methodology that was implemented to create the original dataset is based on systematic screening of various official and unofficial sources that are publicly accessible and contain information on individual Chinese development projects (Strange et al. 2017b). For each project record, we reviewed these underlying sources to double-check existing dates and fill in missing information. In addition, a team of research assistants performed additional English- and Chinese-language internet searches that targeted individual projects that had missing information on commitment, start, and end dates. This was done by using Chinese project titles and the China-based search engine Baidu. To do so, we first translated existing project names into the Chinese language and then searched for the recipient country in combination with a basic project description, name, and year. If the results of this searching were insufficient to find exact project events, we used a more fine-grained approach based on project-specific keywords or the names of the actors potentially involved in the project, such as the respective Chinese

ambassador or the relevant construction contractor. Appendix A (apps. A–D are available online) provides the full codebook that we used to create these data.

This data-gathering effort significantly improved the coverage and quality of project-event dates: we uncovered a total of 3,998 dates. Our study is the first to provide exact commitment dates for Chinese development projects and substantially increased the coverage of start and end dates compared with previous work (for a comparison, see fig. B1; figs. B1–B8 are available online). Of the 3,485 projects included in Dreher et al. (2021b) and Dreher et al. (2022), 429 now contain a commitment date (12%), 1,771 a start date (51%), and 1,798 an end date (52%).²²

These project-event dates allow us to estimate short-term effects of Chinese development projects on public approval of the Chinese government. They also enable us to improve temporal precision in estimates of the more longer-term effects because we can rely on exact project end dates rather than rough estimates based on their commitment year. These data can also benefit future research by enabling researchers to carry out subannual analyses of Chinese development finance. Indeed, AidData's ongoing efforts to track China's global development finance now make use of a project-event-coding protocol similar to the one we developed for this study.

Our event study utilizes these newly collected data by focusing on individuals interviewed just before and after a project-related event. We rely on binary variables indicating these events. In our study of longer-term effects, we rely on the completion date variable exclusively and use the number of Chinese development projects completed in a year as our main variable of interest. We prefer this variable over the (logged) monetary amount of development funding (in USD) to a particular subnational region because 44% of the projects lack information on the monetary values (see Dreher et al. 2021b). The use of project counts circumvents this problem; however, we repeat the analysis using monetary amounts for comparison. In addition, we make use of monetary amounts to investigate the effects of large projects (which we define as those of US\$1 million or above) in separate regressions.

²² Of the 2,183 geocoded projects, 286 projects (13%) now contain information on commitment dates, 1,123 projects (51%) contain information on start dates, and 1,136 projects (52%) contain information on end dates. Commitment dates were considerably more difficult to identify by using our open-source search method. This was in part because online sources appeared less likely to explicitly publish such dates, whereas start and end dates were more common. Similarly, sources did not refer to commitment dates in uniform ways or language (relative to start and end dates, which were more clearly identifiable in these sources).

B. Public Approval of the Chinese Government

We pair our fine-grained Chinese project information with respondent-level data on public opinion from the Gallup World Poll (GWP; Gallup 2018). To the best of our knowledge, the GWP is the most systematic collection of worldwide public opinion data. It covers worldwide annual data since 2006 and includes repeated cross-sectional data for more than 1.5 million individuals—on average 115,000 individuals per interview-year.²³ Each country-wave of the GWP is tethered to a specific period of time when the survey was completed. Individuals are tagged with location variables, which we matched to ADM1 regions.²⁴ This allows us to examine respondent-level data for a maximum of 126 countries and 2,025 ADM1 regions.

GWP data are probability-based and nationally representative of the resident population of 15 years and older—with only a few exceptions due to staff safety concerns and scarcely populated or poorly accessible areas. Questions are standardized around the world for all respondents and asked in the respective national language. GWP interviewers conduct surveys by telephone where telephone coverage exceeds 80% by using random-digit dialing or nationally representative telephone number lists. In all other regions, they conduct face-to-face interviews on the basis of random-routes procedures at different times of the day.²⁵ A typical survey collects data from 1,000 individuals, varying with country population size.²⁶ GWP data are widely used in economic research (Deaton 2008; Bjørnskov 2010; Kahneman and Deaton 2010; Stevenson and Wolfers 2013; Bertoli and Ruysen 2018; Deaton 2018; Guriev, Melnikov, and Zhuravskaya 2022).

While GWP contains several questions that potentially relate to China's soft power, only the following question clearly relates to popular attitudes toward

²³ Figure B2 shows the number of individuals included per year, as well as the months and days of interview.

²⁴ The Gallup World Poll provides within-country geographic variables indicating the subnational region the respondent lives in (named REGION_XXX). We matched this variable with ADM1 shapefile names from the Database of Global Administrative Area (GADM). We use GADM version 2.8 to map the administrative areas of all countries. We successfully mapped provinces from 126 out of the total 140 developing countries in the GWP. To be precise, we include all GWP countries apart from those that belong to the OECD's Development Assistance Committee, which is the group of established donor countries. From these countries, we mapped 2,025 out of a total of 2,280 provinces. If countries or provinces could not be mapped, this was either because the GWP used a spatial identifier that was above the unit of ADM1 regions (but below the country level) or because names of GADM 2.8 units and the spatial identifier did not match. For the full list of countries included in our analysis, see table B1 (tables B1–D9 are available online).

²⁵ Face-to-face interviews usually take about 1 hour; telephone interviews take 30 minutes.

²⁶ For more information on the GWP survey method, see <https://news.gallup.com/poll/105226/world-poll-methodology.aspx>.

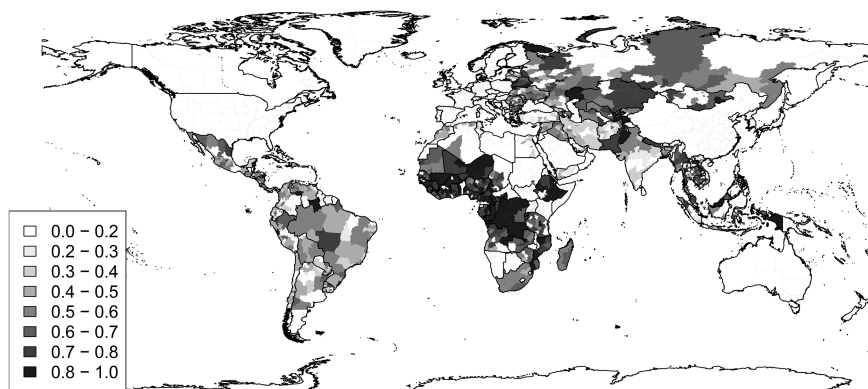


Figure 2. Average support of the Chinese government by province, 2006–17. Shown is the average share of interviewed individuals that approve of the Chinese government by province (ADM1 region) over the sample period (2006–17). Source: Data from Gallup (2018).

China's government and has significant coverage (with 890,000 recorded responses) over time and space: "Do you approve or disapprove of the job performance of the leadership of China?"²⁷ For our empirical analysis below, we recode this variable as a binary indicator that takes the value of 1 if the respondent approves of China's leadership.²⁸ Sixty-three percent of the respondents approve of China's leadership. For comparison, the fraction of respondents who approve of their national government is 54%. The world map in figure 2 visualizes the mean approval rates for the Chinese government over the sample period by ADM1 region. The country with the highest average approval rate is Mali (91%), followed by the Central African Republic (91%) and Guinea (87%). Kosovo (12%), Puerto Rico (16%), and Croatia (25%) have the lowest average approval rates. As can be seen in figure 3, approval rates peaked in 2006 and bottomed out in 2013 but also show considerable differences across world regions. Overall, average approval rates across countries show a decreasing trend over time in our study period.²⁹

III. The Short Run: Individual-Level Event Study

Our goal is to estimate the effects of Chinese development projects on public support for the Chinese government. However, isolating these potential effects

²⁷ The question with the second-best coverage is the following: "In general, what opinion do you have of the following nations? China." However, the coverage of this question is poor with less than 54,000 observations. The correlation between those individuals expressing a positive opinion on the job performance of the Chinese government (our baseline question) and those expressing a positive opinion on China as a nation is 0.36. We use the latter below in robustness tests.

²⁸ We code responses indicating "Don't know" as missing.

²⁹ We give all countries equal weight when calculating the average.

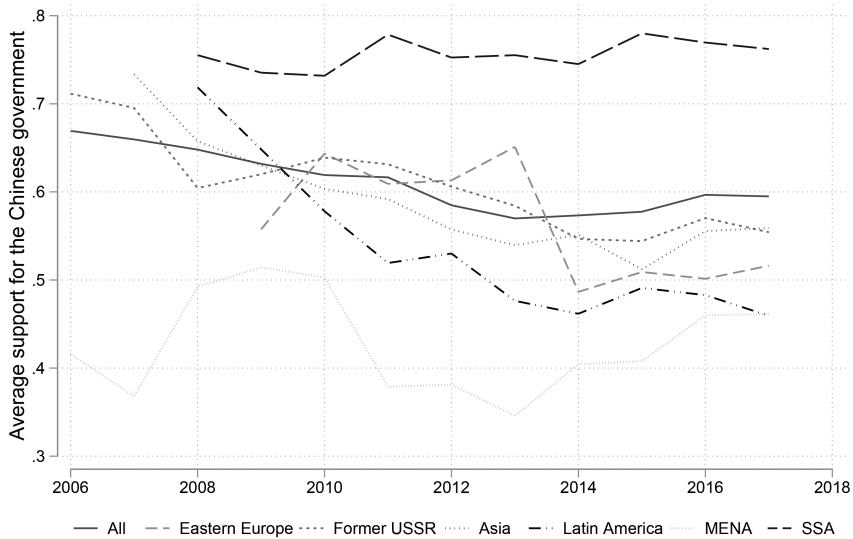


Figure 3. Support for the Chinese government by world region over time, 2006–17. The average approval of the Chinese government by world region over the 2006–17 period is displayed. We first create country means and then world-region averages. USSR = Union of Soviet Socialist Republics; MENA = Middle East and North Africa; SSA = sub-Saharan Africa. Source: Data from Gallup (2018).

is difficult because individuals are exposed to many potentially confounding factors. For example, any other interactions with Chinese state or nonstate actors—whether direct or indirect—could affect individuals’ opinion toward the Chinese government. Many individuals in developing countries regularly observe and interact with Chinese markets, shops, goods, and services, Chinese state-owned and private businesses and investors, managers and workers of Chinese companies, and news coverage provided by local outlets, international media agencies, and Chinese state-run media (Fung et al. 2021).

Given that public opinion toward China and the Chinese government is likely based on composite assessments of these diverse observations and experiences, isolating the effects of Chinese-government-financed development projects is key for our empirical strategy. We address this challenge in two ways. In this section, we use narrow time windows around project commitment, start, and end dates to isolate potential short-term effects of Chinese development projects on approval of the Chinese government. In the next section, we use an instrumental variables approach that relies on exogenous variation in the supply of Chinese development projects to estimate longer-term effects.

A. Empirical Strategy

We carry out an individual-level analysis of short-term effects of Chinese project events. This approach exploits the staggered rollout of the GWP, which specifies the exact calendar days on which respondents are interviewed. All individuals within a country wave are interviewed on days within a time window of about 4 weeks.³⁰ Our new data on project-level event dates allow us to identify GWP country-survey waves that fall in windows around these events. Project events include the three types introduced in the previous section: (1) the commitment of new projects, which are typically announced at bilateral meetings that attract public attention; (2) the start of project implementation, which is visible on the ground and often accompanied by a groundbreaking or foundation-stone-laying ceremony; and (3) project completion, which is often covered by the media and accompanied by a ribbon-cutting ceremony, particularly for larger infrastructure projects. In total, 29,331 individuals in 35 countries and 420 provinces were exposed to 45 projects totalling 53 project events that fall into the respective survey windows over the 2006–17 period.³¹

These 53 project events are covered at random on the basis of whether or not they fall within the windows around GWP interview dates. Both project events and GWP interviews occur throughout the year and cover all weeks of the month and all days of the week (see again figs. B1 and B2). Table B2 tests whether these 45 projects differ in their main characteristics from (i) other projects where information on event dates is also available and (ii) the entire sample of projects (independent of whether information on dates is available). It demonstrates that these projects are representative of all projects in the dataset that have information on commitment, start, and end dates, respectively (cols. 1–3), and of the entire sample of projects (cols. 4–6). These projects do not substantially differ from those without dates across a wide range of project characteristics.³² The subsample of projects covers a wide range of activities, including the construction and expansion of airports, roads, and railways, the dispatching of peacekeeping missions and medical teams, and the construction of primary schools, Confucius Institutes, and government buildings (see tables B3–B5).

³⁰ The mean number is 26 days with a standard deviation of 19 days.

³¹ The number of events exceeds the number of projects because eight events are included twice, with both their start and end dates falling within an interview period.

³² Commitment and start dates are significantly more available for more recent years, which is unsurprising because it should be easier to gather information on more recent events where online media coverage should be better. We capture this by the inclusion of fixed effects in the analyses below. There is also some evidence that commitment dates are more (less) often available for projects in the production sector (ODA projects) and that start dates are more often available for economic infrastructure projects.

Using this subset of projects and project events, we compare individuals interviewed just before a given project event with those interviewed just after it. Our identifying assumption is that the timing of Chinese project events—that is, commitment, start, and end dates—is independent of the timing of the GWP interviews. This is plausibly the case. There is no obvious reason why the interview dates of a US-based survey enumeration firm would systematically be related to the timing of Chinese project events. Nevertheless, we take several measures to ensure that our control group (individuals interviewed just before a project event) is comparable with our treatment group (individuals interviewed just after a project event). First, we compare only individuals interviewed in the same province and year, because we control for province-year level confounding factors through province-year fixed effects. Second, we limit the sample to include only individuals interviewed 30 days before and after the event to further mitigate the probability that individual opinion is driven by other events.³³ Third, we add survey-level control variables to ensure that the effect is not driven by any underlying implementation of the GWP survey in a specific wave. On the basis of these measures, it is arguably random if an individual is interviewed before or after a project event, which allows us to interpret our results in a causal manner.

Specifically, we estimate the following equation:

$$\text{Support}_{ipcdy} = \beta \text{post}_{icdy} + \gamma X_{ipcdy} + \delta S_d + \zeta_{pcy} + \epsilon_{ipcdy}, \quad (1)$$

where Support_{ipcdy} is a binary variable that takes a value of 1 if individual i living in province p of country c interviewed on day d in year y approves of China's government (as introduced in the previous section), and post_{icdy} indicates whether or not an individual has been interviewed within 30 days after a project event in the country—either commitment, start of the implementation period, or completion.³⁴ The term X_{ipcdy} represents individual-level control variables, which include a binary variable if the respondent is female, the respondent's age in years and its square, an education indicator, and an urban area indicator.³⁵

³³ We choose 30 days because it roughly corresponds to 1 month, but we also test robustness to other event windows.

³⁴ To be conservative, we include the treatment day in the control group. By doing so, we assume that events take some time to unfold. Results are robust to excluding the event day from the analysis instead.

³⁵ The education variable takes a value of 1 if the respondent has 1–8 years of schooling, a value of 2 for 8–15 years, and a value of 3 for 15 years or more. The urban area indicator is 1 if the respondent lives in a rural area or village, 2 for a small town, 3 for the suburb of a large city, and 4 for a large city. Panel A of table B6 provides descriptive statistics of all variables employed in the individual-level analysis.

The term S_d represents survey-level control variables, including binary variables for the day of the week and an indicator counting the days of the GWP survey windows.³⁶ The term ζ_{py} denotes province-year fixed effects. Finally, ϵ_{ipcdy} is the error term. We cluster standard errors at the level of the treatment, which is the country $\times$ interview date. The term β is the coefficient of interest that shows the average effect of Chinese projects on support.³⁷

Our coefficient of interest β will likely represent a lower bound of the true effect. First, it is likely that public opinion in the preproject period is affected by other, earlier projects, which biases our coefficients toward zero.³⁸ Second, it is likely that news related to project events disseminates in the days just prior to an event. However, it appears unlikely that within such short periods of time, the nature of disseminated news changes systematically in one direction. If reporting prior to the event exists at all, it seems likely that reports that are positive (negative) about China in the days before commitment will also be positive (negative) after commitment. We would then attribute parts of the effect to the control group, which would bias our estimated effect downward. For example, consider Kenya's Standard Gauge Railway (SGR). The SGR was officially committed to on May 11, 2014, but became known through the media earlier when the China Road and Bridge Corporation started negotiating with Kenya's government. The *Daily Nation*, one of Kenya's most influential newspapers, started publishing articles that made reference to the proposed SGR by early 2014. While the vast majority of media coverage occurred well after the project's commitment date, five articles were published before commitment, but only one of those within 30 days before the official commitment date. Given that this article is assigned to the control group, it could bias our estimate downward. We will thus interpret our results as lower-bound estimates.

Estimating the effects of development projects on Chinese government approval for all individuals in a given country is suitable to the extent that (countrywide) reporting about projects drives attitudes rather than local news or personal experiences on the ground. For China and other donor governments, country-level attitudes are likely more important than local attitudes in the pursuit of soft power. However, by additionally estimating provincial effects,

³⁶ When we further add fixed effects for each month, results are virtually unchanged (see table C1).

³⁷ Our results are robust against a variety of different levels of clustering standard errors, including two-way clustering for country and interview date, country and interview wave, province and interview date, province and interview wave, country, province, wave, and date. When we cluster for country $\times$ interview wave or wave, our results remain similar but are slightly weaker in terms of statistical significance.

³⁸ Results are robust when leaving out those projects where another project was committed to, started, or completed in the 6 months prior to the project event.

we can also account for local experiences. While our main analysis focuses on project events anywhere in a country, we also estimate a variant of equation (1) in which we add an interaction term of post_{icdy} with a binary indicator for projects that are placed in the province of the person interviewed, testing whether opinions of individuals who have been interviewed within 30 days after a project event in the province differ from those interviewed elsewhere in the recipient country. We thus also present results that compare individuals of the same province interviewed before a project date with those interviewed thereafter.

B. Micro-Level Results

On the basis of the event-study approach of equation (1), table 1 shows our main results for the short-run effect of Chinese development projects on public opinion toward China's government. Column 1 of panel A investigates the effect of project events for all projects in our sample—small and large ones alike and independent of whether or not they are concessional. The coefficient on “Post” shows that there is no significant difference in opinion about the Chinese government across individuals interviewed after as compared with before a project event. In columns 2–4 of panel A, we investigate these events separately by event type. This reduces the number of projects included in the analysis substantially. Column 2 shows the average effect of 9 committed projects, column 3 that of 19 projects that start the implementation phase, and column 4 that of 20 completed projects. The results show that only project completion significantly affects public opinion, at the 10% level of significance. Being interviewed after the project end date increases Chinese government approval by 3.0 percentage points.³⁹ This effect is sizable in light of the sample mean of 70% and given that we likely obtain lower-bound estimates for the reasons outlined above.

We offer four potential explanations for this strong result: First, completed projects are often touted by recipient and donor governments as visible achievements. Project completion dates erase any uncertainty about whether a project will actually reach completion. Second, the completion of a project ends negative spillovers or externalities directly stemming from the construction or other elements of project implementation. By contrast, when individuals initially hear of a project commitment, they might not expect positive effects from the project

³⁹ We have tested heterogeneity on a number of dimensions: We find no evidence that individuals with better education or access to information evaluate projects differently. Nor do we find lagged approval of China's government, one's own government, or the US government to affect how China's projects affect approval. While we do not report these results in a table to reduce clutter, they are available on request.

TABLE 1
CHINESE PROJECTS AND GOVERNMENT SUPPORT, EVENT-STUDY RESULTS

	All (1)	Commitment (2)	Start (3)	End (4)
A. All Projects				
Post	.0108 (.0121)	-.0509 (.0373)	.0225 (.0154)	.0303* (.0156)
Observations	29,331	5,610	15,362	15,465
Number of countries	35	9	19	20
Number of provinces	420	128	185	247
Number of projects	53	10	21	22
Province-year FE	✓	✓	✓	✓
B. ODA Projects				
Post	.0228 (.0145)	-.0421 (.0812)	.0462** (.0183)	.0412** (.0171)
Observations	19,017	1,744	10,528	12,226
Number of countries	22	3	14	15
Number of provinces	265	58	125	171
Number of projects	35	3	15	17
Province-year FE	✓	✓	✓	✓
C. Large Projects				
Post	.0243 (.0185)	.0347 (.0561)	.0413* (.0249)	.0574** (.0272)
Observations	13,783	2,169	5,619	7,865
Number of countries	19	5	7	11
Number of provinces	254	61	62	161
Number of projects	24	5	8	11
Province-year FE	✓	✓	✓	✓

Note. Dependent variable is binary and indicates whether or not an interviewed individual approves of the Chinese government on the basis of the question "Do you approve or disapprove of the job performance of the leadership of China?" The sample is restricted to individuals interviewed within 30 days before or after ("Post") a project-related event date. Panel A shows results for the full sample of projects that contain information on the specific event date and fall into the interview window. Panel B includes only "ODA-like" projects. Panel C includes only large projects, i.e., those with a size of US\$1 million or above. All specifications include individual-level and survey controls as well as province-year fixed effects. Standard errors (in parentheses) are clustered by country-date. FE = fixed effects.

* $p < .10$.

** $p < .05$.

and only realize such effects after project completion. Third, individuals might learn about where and what types of projects China's government has provided only after a project's completion. Finally, there is a fourth, more technical, explanation. As can be seen in figure B5, approval of China's government is above average on some days prior to the respective commitment and start dates, which makes it less likely we observe positive approval effects of these project events. To the contrary, there are no significant differences from average approval rates prior to project completion dates, so there is no downward bias for these dates.

A skeptical reader might worry that the presence of pretrends could render our results spurious. To guard us against such possibility, we draw 60-day periods from the GWP data at random. We do this 22 times, because this corresponds to the number of project end dates in our sample. We then draw one random date from each of the random periods and use those as placebo treatments in our regression analyses. We repeat this procedure 999 times.

Figure B6 shows the results of this randomization inference test based on 999 Monte Carlo replications. As can be seen, the results follow a normal distribution centered around zero. The p -value from an exact Fisher test shows that the absolute value of the t -statistic in the simulated data exceeds that of the original t -statistic in 8.1% of the regressions. These results indicate that spurious trends cannot explain the significant results from our event study.

To win the approval of the general public, donors often emphasize that their projects represent acts of generosity (Dietrich, Hyde, and Winters 2019). Individual projects vary significantly in their financial terms and degree of generosity, and most donors provide a combination of highly concessional ODA and less concessional (or nonconcessional) OOF. Panel B of table 1 reports the effects of highly concessional projects only. At the 5% level of significance, we again find that public opinion of the Chinese government improves when such projects are completed (col. 4). Individuals interviewed within 30 days of the completion of a Chinese ODA project approve of the Chinese government by 4.1 percentage points more than individuals interviewed in the prior 30-day period.⁴⁰ It is not surprising that when compared with projects financed at market or near-market rates, projects financed on highly concessional terms more effectively increase public support for donor governments—due to either perceived generosity or competence.

Development projects also differ substantially in terms of their physical and financial size. We expect larger projects to produce greater public opinion effects not only because they signal more generosity but also because they possess higher levels of visibility within recipient countries. For one, larger projects have a higher probability that individuals within a developing country accurately attribute them to the donor country. Such attribution should not be taken for granted. Indeed, experimental evidence indicates that citizens often do not know which foreign governments finance specific development projects (Baldwin and Winters 2020). What is more, compared with smaller projects, larger aid projects are often more visible in terms of both their physical and media presence.

⁴⁰ These results are robust to decreasing the bandwidth to 20 days and to increasing it to 40 or 60 days (table C2).

Earlier research indeed suggests that development-project visibility matters for both recipient- and donor-country governments.⁴¹ Panel C of table 1 includes only large projects, which we define as those with commitment values of US\$1 million or more. As expected, the observed effect for large projects (of 5.7 percentage points) is stronger than for all projects. Moreover, approval of the Chinese government increases when concessional and large projects commence, at the 5% level and 10% level, respectively (see col. 3).⁴²

Next, we test whether the effect of project events differs for respondents from the province where the project is located compared with the effect for all individuals interviewed in the recipient country. To this end, we include the interaction of the treatment indicator with a binary variable that indicates a project is from the same province as the interviewee.⁴³ The results show only two significant coefficients on this province-specific indicator in our 12 specifications (see table C4). First, the commitment of one additional project increases public approval for the Chinese government (among people interviewed in the project province) by 15.1 additional percentage points—compared with the effect that is observed among all interviewees outside the project province (col. 2). Second, albeit smaller in size, we also find that project completion events trigger a larger response for ODA projects (col. 8). Because all other interaction terms are insignificant at conventional levels, it seems that the overall effects reported above are not driven by provincial- or local-level opinion effects, but rather by project-specific news or information that is accessible to citizens nationwide. Given that our event-study approach measures short-term effects where experiences with project outcomes should differ less across individuals, this result makes intuitive sense. To capture potentially heterogeneous individual experiences with project outcomes, we need to look at a longer time frame. We do this in the next section.

As an additional robustness test, we turn our attention to China-related events that might coincide with project events and potentially bias our estimates. One possibility is that high-ranking official visits by Chinese leaders take place around the same time as project events. We would then falsely attribute the

⁴¹ For recipients, completing highly visible projects with tangible site locations can benefit incumbent host country leaders' reelections (Marx 2018). Highly visible projects can also potentially send clearer signals of political support between donor and host governments (Strange 2024).

⁴² For completeness, we also tested the effects of projects with funding below US\$1 million and of OOF-like projects. Table C3 shows that 9 out of 10 coefficients are not significant at conventional levels. The exception is the effect of commitment-related news on small projects, which is negative at the 10% level of significance. However, this latter finding should not be overinterpreted because it relies on only four projects.

⁴³ Note that we include province fixed effects that capture the constituent term of the interaction.

effects of those visits to project events.⁴⁴ To mitigate such concerns, we investigate state visits by the Chinese presidents that occur around project dates. Using data from Wang and Stone (2023), we find that only five leader visits fall within the 30-day windows around project events (3 commitment dates, 1 start date, 1 end date). When we exclude these five events, our results are qualitatively the same, as we show in table C5. If anything, they become stronger. This makes us confident that our results are not significantly biased by high-ranking diplomatic visits that happen to be at the same time. While we cannot fully rule out that other China-related events might affect our results, we deem it unlikely that these would bias our results in a systematic manner given that we observe no bias in case of state visits, which are clearly exceptional events.

Finally, we test whether we come to the same conclusions when we repeat the analysis by using an alternative measure of China's soft power. Responses for other questions in the GWP potentially related to Chinese soft power are, however, of very limited availability. We test whether our results hold when we use a respondent's general opinion on China as outcome variable, which is the most populated variable after our main outcome. The underlying question is the following: "In general, what opinion do you have on the following nations? China," which we recode to 1 if the opinion is favorable and 0 if the opinion is unfavorable. The question is answered by 60,000 individuals (compared with 900,000 that answered our baseline question whether or not they approve the job performance of the Chinese government). Despite this low number, our main results hold when using the opinion of China as dependent variable (table C6).

IV. The Longer Run: Macro-Level Analysis

A. Empirical Strategy

A disadvantage of the above event study is that we can include only a small set of projects that are committed, started, or completed during GWP survey windows in a particular country. This also implies that we rely on only a small portion of interviews from the GWP. Moreover, while the event study allows us to identify causal effects, we can test effects only shortly after an event has happened and cannot analyze potential longer-term effects. We therefore reanalyze our dataset at the annual level in order to exploit the full range of projects. We estimate two sets of equations, one at the province level and one at the country

⁴⁴ In this regard, Goldsmith, Horiuchi, and Matush (2021) find that leader visits can increase public approval among foreign citizens, indicating that public diplomacy has the potential to affect soft-power variables.

level.⁴⁵ The province-level regressions allow us to analyze more granular potential effects and are instrumental for our identification strategy at the country level, as we explain below. The country-level regressions capture the public opinion effects of Chinese-government-financed projects on an entire country regardless of individuals' physical proximity or even direct exposure to projects. From the perspective of a donor government interested in maximizing its soft power, the country level is arguably more important. Starting with the province level, we estimate the following regression equation:

$$\text{Support}_{cpy} = \beta \text{ChineseProjects}_{cpy-1} + \sum_j \delta_j X_{cpy}^j + \zeta_{cp} + \eta_{cy} + \epsilon_{cpy}, \quad (2)$$

where Support_{cpy} is the share of individuals that approve of China's government in province p of country c interviewed in year y .⁴⁶ The term $\text{ChineseProjects}_{cpy-1}$ denotes the number of Chinese development projects completed in a province in the previous year.⁴⁷ The term X_{cpy}^j represents our individual-level control variables introduced in the previous section and averaged at the province level; ζ_{cp} represents province fixed effects; and η_{cy} denotes fixed effects for country-years. The inclusion of country-year fixed effects implies that any effects of development projects that affect public perception equally across provinces are netted out so that the effect of projects at the province level has to be interpreted relative to this average effect. Again, ϵ_{cpy} is the error term; we cluster standard errors at the level of countries.⁴⁸

Estimates obtained from equation (2) likely suffer from endogeneity bias because projects are not randomly distributed over time and space. For example, reverse causality might bias our results if development projects are deliberately placed in regions to increase support at times when the Chinese government is unpopular (Asmus et al., forthcoming). Conversely, provinces with individuals who possess relatively favorable opinions of China in a certain year could obtain more projects if China believes that public support is conducive to project success or if provincial leaders maintain close ties to the Chinese government. For instance, Blair, Marty, and Roessler (2022) find that Chinese development projects (in Africa) are often situated in regions where public opinion about

⁴⁵ Panels B and C of table B6 show the descriptive statistics on the province and country levels.

⁴⁶ We make use of sampling weights by GWP when aggregating individual responses to the province or country level. Although we believe that the GWP weights are preferable, table D2 presents our main regression results where we weight provinces with the number of individuals surveyed as share of total survey respondents in the country-year wave. Results are very similar.

⁴⁷ Given that this specification aims to test the longer-run effects of Chinese development projects, completion dates are arguably most appropriate.

⁴⁸ Results are robust when we cluster at the level of (i) country-year and provinces, (ii) provinces, and (iii) province-year. When we cluster at the level of provinces, however, the first-stage F -statistic is lower (8.6).

China is more favorable relative to other locations. Jones (2018) similarly shows that public opinion about China in Africa is more favorable in the home regions of political leaders in aid-receiving countries, which tend to receive a higher share of Chinese-funded projects (Dreher et al. 2019a).

We address the endogeneity of development projects using a two-stage least squares (2SLS) regression, replacing $\text{ChineseProjects}_{cyy-1}$ in equation (2) with $\text{ChineseProjects}_{cyy-1}$, based on the following first-stage regression:

$$\text{Chinese Projects}_{cyy-1} = \tilde{\beta}(\text{Input}_{y-3} \times \mu_{cp}) + \sum_j \tilde{\delta}_j \tilde{X}_{cyy-1}^j + \tilde{\xi}_{cp} + \tilde{\eta}_{cy-1} + \tilde{\epsilon}_{cyy-1}, \quad (3)$$

where we rely on a two-part instrumental variable for Chinese development projects introduced in Bluhm et al. (2025). First, relative to domestic demand, China overproduces cement, iron, steel, and other construction inputs. China's government instructs Chinese commercial actors to offload excess production in foreign markets. Bluhm et al. (2025) show that excess supplies of physical project inputs in China leads to increases in Chinese development projects abroad. More precisely, we use a proxy for China's supply of potential project inputs that includes (logged) production of six raw materials: aluminum, cement, iron, steel (all measured in 10,000 tons), glass (measured in 10,000 weight cases), and timber (10,000 cubic meters). As in Bluhm et al. (2025), from these six inputs, we draw the first factor using factor analysis and detrend the resulting time series.

The first factor of Chinese raw material production captures annual fluctuations in (potential) Chinese project inputs but does not include spatial variation in terms of the countries and provinces that receive the additional financing from China. The second part of the instrument, also introduced in Bluhm et al. (2025), is based on the intuition that locations that receive Chinese development projects more frequently will be more affected by changes in China's overall supply of project inputs. We thus interact the measure for China's project inputs described above with a province's probability of receiving Chinese development projects, which we proxy with the share of years during 2000–2014 in which at least one project was completed in a given country or province. Following a growing body of research on aid effectiveness, we thus interact a variable that varies exclusively over time with a variable that varies over space (ADM1 regions), resulting in an instrumental variable that varies in both dimensions.⁴⁹

⁴⁹ See Werker, Ahmed, and Cohen (2009), Dreher et al. (2019), Lang (2021), and Gehring, Kaplan, and Wong (2022), among others. Dreher et al. (2021b) propose a second instrument—the change in

In equation (3), Input_{t-3} is the (logged and detrended) factor of the production levels of the six physical project inputs, and μ_{cp} is the province-specific probability of receiving Chinese development projects.⁵⁰ The term ξ_{cp} represents province fixed effects, and η_{cy-1} represents fixed effects for country-years, capturing the production of Chinese project inputs. On average, projects are completed within 2 years after commitment.⁵¹ The time from commitment to project completion, however, largely varies by project sector, so using the time of project completion instead of commitment in combination with a 2-year lag allows us to measure the effect of development projects more precisely. Allowing one additional year for the projects to take effect, we lag completed projects by 1 year. We lag input factor production by two additional years so that it is measured in $y - 3$ relative to the year of an interview.

The intuition behind our identification approach resembles a difference-in-differences design. We investigate a differential effect of surplus Chinese project inputs on public opinion in regions with a high probability versus those with a low probability of receiving Chinese development finance. The identifying assumption is that other than the effect of Chinese development finance on public opinion, public opinion in provinces that have differing probabilities of receiving development finance from the Chinese government will not be differentially affected by changes in China's production of physical project inputs, after controlling for province-year fixed effects and country-year fixed effects and the other variables in the model. As in any difference-in-differences setting, we rely on a conditionally exogenous treatment and the assumption of parallel trends across groups. In controlling for country-year fixed effects, Chinese production volumes of project inputs cannot be correlated with the error term and are thus (conditionally) exogenous to China's provision of international development projects. For different trends to exist, these trends across the treatment and control groups—provinces with a high probability and provinces with a low probability of receiving development finance from China—would have to vary in tandem with period-to-period changes in the production of project inputs.

China's currency reserves in a year interacted with the probability of receiving Chinese development finance. In our comparably short sample, currency reserves and the production of China's construction materials are highly correlated (0.85) so that little information is added by including the second instrument. When we do, results are almost identical, with substantially lower first-stage Kleibergen-Paap F -statistics.

⁵⁰ Specifically, we calculate this as the share of years in which at least one Chinese development project was completed over the entire time these data are available as follows: $\mu_{cp} = 1/15 \sum_{t=1}^{15} \mu_{cpt}$, where μ_{cpt} is a binary variable indicating whether a development project has been completed in a given province in year t .

⁵¹ Figure B3 visualizes the average time between project event dates; see fig. B4 for sector-specific results.

We assess the validity of this approach in several steps following Bluhm et al. (2025). First, the identifying assumptions of the above approach could be violated if fluctuations in China's production in project inputs had different effects on "regular" and "irregular" recipient provinces' probabilities of receiving Chinese development projects, and if such effects had unique consequences for public opinion toward China's government across these provinces. In other words, this approach provides plausibly exogenous timing of the intervention, but its validity still rests on parallel pretreatment trends across regular and irregular provinces that receive high and low levels of Chinese development finance. In figure B7, panels A and B display raw production volumes of the six input materials—both in levels and linearly detrended. Panels C and D report the first factor of the production volumes that we extract via factor analysis (again, both in levels and detrended). Panels E and F plot variation in the number of projects completed and public opinion for districts with above- or below-median probabilities to receive development finance from China. Overall, there is minimal concern that the parallel-trends assumption is violated. We observe a global upward trend in material production. After purging the linear trend, the series trends upward until 2010 and then turns downward. This trend is reflected in China's global provision of development projects over the same period, as we would expect, because material production serves as input into Chinese development projects. Moreover, the probability-specific trends for projects and public opinion appear generally parallel for the provinces with above- or below-median probabilities to receive Chinese projects. There is no apparent nonlinear trend that resembles the trend of the detrended first factor of Chinese domestic input production in one of the two groups (above- or below-median probability to receive development finance) more than the other (Christian and Barrett 2024).

Second, to obviate the assumption of parallel pretreatment trends, we allow for correlated random trends. As Bluhm et al. (2025) point out, the identifying assumption is $\text{Cov}(\text{Input}_{y-3} \times \mu_{cp}, \epsilon_{cpt}) = 0$, conditional on the set of control variables and fixed effects. This approach minimizes potential confounding factors. One possibility is that the detrended input series might be correlated with production volumes or prices of other commodities apart from our six input materials. If so, and if the time-varying effects of these variables on public opinion were equal across all regions in a country, then detrending the raw series and including country-year fixed effects would capture such effects. If the effects were instead linear but different across provinces, then province fixed effects would capture them. Finally, if time-varying effects were both nonlinear and different across provinces, one would need to account for such potential shocks. We address this possibility in the robustness tests in appendix D.2 (see table D8).

Third, a separate concern is whether China's domestic production of physical project inputs could be correlated with China's trade or foreign direct investment volumes (Dreher et al. 2021a; Bluhm et al. 2025). China's share of world manufacturing value-added has risen steadily since 2000, and this coincided with a large demand shock for raw materials (Autor, Dorn, and Hanson 2016). What is more, frequent recipients of Chinese development projects might also be popular destinations for Chinese foreign direct investment projects or have closer trade ties with China. If so, differences in public opinion across provinces in developing countries might actually be due to trade or foreign direct investment rather than Chinese-financed development projects. To investigate this possibility, we conduct robustness tests that control for annual volumes of exports to China, imports from China, and Chinese foreign direct investment. We interact these volumes with a set of variables that makes it more or less likely that a province will be affected by variation in China's overall trade or investment (app. D.2; see table D9).

Our identification strategy is related to a growing literature that employs shift-share instrumental variables. Such instruments are often constructed as macro-level shocks to a variety of industries that have varying degrees of local exposure. Within this setting, one can achieve identification in two ways by using alternative assumptions. In the first approach, if local industry shares are plausibly exogenous, they can be taken as instruments (Goldsmith-Pinkham, Sorkin, and Swift 2020). In the second approach, even when variation in local exposures may be endogenous, one can alternatively stake identification on exogenous variation in time-series shocks.⁵² Our approach is somewhat different because we rely on endogenous exposure to a single (and perhaps endogenous) shock. Rather than arguing that the shock is necessarily exogenous, we rely on alternative assumptions discussed above.⁵³

⁵² For the panel case, Borusyak, Hull, and Jaravel (2022) show that the estimator is consistent when the covariance between the detrended input series and a weighted average of the within-location time variation in unobserved factors affecting public opinion approaches zero in large samples. This is likely with reasonably large T , combined with a set of fixed effects, and can be supported by including proxies for the remaining unobserved variation (Bluhm et al. 2025).

⁵³ We conduct several tests suggested by Christian and Barrett (2024) to probe the validity of these assumptions. In addition to visually examining trends in fig. B7, we conduct a randomization inference test where we reassign the number of projects and the corresponding instrumental variable to different countries and years in the sample, just as we did for the event study in fig. B6. As can be seen from the results of 999 regressions reported in fig. B8, coefficient estimates are concentrated around zero. According to an exact Fisher test, the coefficient from our main estimate above (indicated by the vertical dashed line) is significantly different from the randomized coefficients (p -value: .078). This

Next, we turn to country-level regressions and employ the same approach used in our province-level regressions shown in equation (2) and equation (3). We aggregate the predicted number of projects in all provinces in a year from equation (3) to the level of countries. We then use these aggregates as instrumental variables for the total number of new projects at the country level.⁵⁴ Alternatively, one might consider to directly estimate 2SLS regressions at the country level. However, the corresponding first-stage Kleibergen-Paap F -statistics are too low (see panel A of table D3).⁵⁵ This is why we use the following first-stage regression equation that relies on the predicted number of projects:

$$\text{ChineseProjects}_{cy-1} = \tilde{\beta} \overline{\text{ChineseProjects}_{cyy-1}} + \sum_j \tilde{\delta}_j \tilde{X}_{cy-1}^j + \tilde{\zeta}_c + \tilde{\eta}_{y-1} + \tilde{\epsilon}_{cy-1}. \quad (4)$$

In the second stage, we then run country-level regressions with individual-level variables averaged at the country level and controlling for fixed effects for countries and years:

$$\text{Support}_{cy} = \beta \overline{\text{ChineseProjects}_{cy-1}} + \sum_j \delta_j X_{cy}^j + \zeta_c + \eta_y + \epsilon_{cy}, \quad (5)$$

where Support_{cy} is the share of individuals that approve of China's government in country c interviewed in year y ; $\overline{\text{ChineseProjects}_{cy-1}}$ denotes the predicted number of projects completed in a country in the previous year; X_{cy}^j represents our j individual-level control variables introduced in the previous section; ζ_c represents country fixed effects; and η_y denotes fixed effects for years. Again, ϵ_{cy} is the error term; we cluster standard errors at the level of countries.

holds when we break the timing structure required for identification and instead randomize the entire time series between countries, years within countries, and countries within years. In short, it is unlikely that any omitted variables correlate with our variables of interest in a way that spuriously produces our main results.

⁵⁴ Note that standard errors are estimated consistently as long as the second-stage error term is not correlated with the instrumental variable of the regression we use to predict the generated instrument (Wooldridge 2010). In our case, the exclusion restriction using province-level probabilities to receive aid rather than probabilities at the country level does not pose additional demands. When we employ wild bootstrap at the second stage, results are unchanged. Authors using a dyadic zero-stage regression to construct their instrumental variable just like we do are Frankel and Romer (1999), Rajan and Subramanian (2008), and Dreher and Langlotz (2020).

⁵⁵ The low power of the instrument compared with the instrument of Dreher et al. (2021b) can be explained by the shorter sample period in our study (2006–17). When we repeat the first-stage regression with the full 2000–2017 sample, the instrument has sufficient power. We show these results using the out-of-sample prediction in panel B of table D3. They are similar to those of our baseline approach reported below.

B. Macro-Level Results

Table 2 presents the results on the longer-term effects of completed development projects, following equations (2)–(5). Columns 1 and 2 show the results for all projects at the level of provinces and countries, respectively. Columns 3 and 4 show the analogous regressions for ODA projects only; columns 5 and 6 show those for large projects. While we calculate probabilities to receive aid for these regressions on the basis of ODA and large projects, respectively, these

TABLE 2
CHINESE PROJECTS AND GOVERNMENT SUPPORT, INSTRUMENTAL VARIABLES RESULTS

	(1)	(2)	(3)	(4)	(5)	(6)
A. OLS						
Chinese projects _{<i>y-1</i>}	-.00380 (.00301)	.00184** (.000852)	-.00390 (.00353)	.00167** (.000741)	-.000135 (.00342)	.00127* (.000669)
B. Reduced Form						
Input × probability _{<i>y-3</i>}	-.139** (.0677)		-.173** (.0721)		-.188* (.0997)	
Σ _{<i>p</i>} Input × probability _{<i>y-3</i>}		.00252** (.00109)		.00233** (.000972)		.00172* (.000893)
C. 2SLS						
Chinese projects _{<i>y-1</i>}	-.0722* (.0389)	.00206** (.000973)	-.0864** (.0409)	.00189** (.000859)	-.0682* (.0400)	.00140* (.000771)
D. First Stage						
Input × probability _{<i>y-3</i>}	1.929*** (.410)		1.998*** (.375)		2.758*** (.719)	
Σ _{<i>p</i>} Input × probability _{<i>y-3</i>}		1.222*** (.0568)		1.230*** (.0522)		1.230*** (.0483)
Level	Province	Country	Province	Country	Province	Country
Project type	All	All	ODA	ODA	Large	Large
Observations	6,296	452	6,296	452	6,296	452
Number of countries	91	90	91	90	91	90
Number of provinces	1,399		1,399		1,399	
Country FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Country-year FE	✓		✓		✓	
Province FE	✓		✓		✓	
F-statistic	22.14	464.4	28.32	555.1	14.72	650.6

Note. Dependent variable in panels A–C is the share of individuals that approve of the Chinese government in a given province/country on the basis of the question “Do you approve or disapprove of the job performance of the leadership of China?” Dependent variable in panel D and variable of interest in panels A–C is the number of Chinese development projects completed in the previous year (Chinese projects_{*y-1*}). Columns with project type “ODA” include only “ODA-like” projects. Columns with project type “Large” include only projects with a size of US\$1 million or above. Columns with level “Province” (“Country”) contain results of regressions at the province-year (country-year) level. All specifications include the control variables age, age squared, gender, education, and urban in addition to the set of fixed effects indicated in the table. Standard errors (in parentheses) are clustered by country. OLS = ordinary least squares.

* $p < .10$.

** $p < .05$.

*** $p < .01$.

probabilities are highly correlated with those to receive any type of development project from China. This violates the exclusion restriction to some extent. While we report these separate results for completeness, we acknowledge that our approach does not allow us to neatly identify separate effects across project types.⁵⁶

Panel A presents the results from OLS regressions that leverage within-country or within-province variation. Consistent with the results from the event study, we find significant and positive (conditional) correlations at the level of countries. However, controlling for country-year fixed effects, we do not find additional effects at the province level. This suggests the reputational benefits of completed development projects are not primarily based on improved conditions near project sites. This pattern suggests the public opinion gains resulting from Chinese development projects are attributable to changing national perceptions rather than the firsthand observations by or experiences of people with direct project exposure.

Panel B in table 2 reports the reduced-form estimates for the same set of regressions. Here we regress average attitudes toward the Chinese government on our instrumental variable (in addition to the fixed effects and control variables). If our identification strategy holds in the presence of an effect of development projects on approval rates, we should also observe strong reduced-form effects. Indeed, there is a sizable and significant effect of the instrumental variable on approval at the level of countries and provinces. At the country level, the coefficient is positive and statistically significant for all projects combined (shown in col. 2), ODA projects (col. 4), and large projects (col. 6) alike. The effect at the province level is negative and statistically significant at conventional levels (as shown in cols. 1, 3, and 5), which suggests that those living in close proximity to Chinese development projects (i.e., those who have more direct project exposure) develop less favorable views of the Chinese government compared with the average effect in the same country and year. This finding is consistent with previous research that provides evidence of higher levels of corruption, political capture, ethnic tension, social protest, and environmental degradation in areas within close proximity of Chinese development projects (Isaksson and Kotsadam 2018a, 2018b; Dreher et al. 2019a; Isaksson 2020; Iacolla et al. 2021; Baehr, BenYishay, and Parks 2023). The effects will be passed through with the same sign if the corresponding first-stage regression

⁵⁶ What is more, the local average treatment effects that we estimate with our input-material-based instruments might capture similar projects in case we use them to predict ODA projects, large projects, or all of them. Very small projects, for example, are less likely to depend on input materials than large ones.

is sufficiently strong and the coefficients on our instrument are positive, as expected.

In panel C of table 2, we present our main results in which we instrument development projects completed 1 year ago with our instrument 2 years earlier as in equation (3). At the country level (cols. 2, 4, and 6), the coefficients increase slightly compared with the corresponding OLS estimates and remain statistically significant at least at the 10% level. Each additional Chinese development project that reaches completion increases public approval for the Chinese government by 0.21 percentage points. While this longer-run effect is smaller than the short-term increase of 3.03 percentage points identified in the previous section, this finding demonstrates that public opinion effects persist over a longer period of time. On balance, it appears that China's development program pays off in terms of soft-power acquisition.⁵⁷

Panel D in table 2 reports our corresponding first-stage regression results. Recall that we instrument the number of countrywide projects with the sum of predicted province-level projects. It is reassuring that the corresponding first-stage Kleibergen-Paap *F*-statistics are all high with values of 464 (all projects), 555 (ODA projects), and 651 (large projects). As expected, we observe a positive relationship between our instrumental variable and the number of development projects completed. A 2 standard deviation increase in "Input" (which is 0.6) creates an additional project in a province hosting multiple projects in 6 years over the 2000–2014 period (the 10th percentile in the sample) but creates only approximately 0.2 projects in a province hosting one project in the sample period (the median over the 2000–2014 period).⁵⁸

At the level of provinces (cols. 1, 3, and 5 of panel C), the coefficients stay negative but increase by an order of magnitude compared with the corresponding OLS estimate (and gain statistical significance at conventional levels). Measurement error, reverse causality, and omitted variables seem to conspire to bias our OLS coefficients upward, therein highlighting the need for instrumentation. Each additional Chinese development project reduces public approval for

⁵⁷ Table D1 provides sector-level results. We discuss those results in detail in app. D.1. We also repeated the analysis with opinion on the Chinese nation (rather than the Chinese government) as alternative measure of soft power. Because this alternative question is answered only by 60,000 individuals (compared with 900,000 that answered our baseline question whether or not they approve the job performance of the Chinese government), we assumed this number to be too low to estimate precise estimates, and indeed this holds true (table D4).

⁵⁸ Projects committed in earlier years can affect today's support of the Chinese government as well. To account for this, table D7 estimates cumulative effects. Specifically, we use moving averages for both project numbers and our instrumental variables of up to 5 years. The coefficient size remains similar but is less precisely estimated with moving averages of several years. We discuss this in greater detail in app. D.2.

the Chinese government by 7.22 percentage points (relative to any country-wide effects that we capture in country-year fixed effects). Reassuringly, the coefficients in the first-stage regression presented in panel D are highly significant, and the first-stage Kleibergen-Paap F -statistics associated with the province-level regressions are considerably larger than the conventional rule-of-thumb value of 10.⁵⁹

Several factors may explain why development projects yield popularity gains for the donor government in recipient countries and why they are less pronounced in the target provinces themselves.⁶⁰ We test possible explanations in the next subsection.

C. Testing Mechanisms

The existing literature suggests several mechanisms that may explain how aid affects public approval of a donor government. We test these mechanisms with country-level regressions following equation (5) in panel A of table 3, where we replace our dependent variable, $\text{Support}_{c,y}$, with potential intermediate outcomes.

First, previous research shows that Chinese development projects lead to short-run economic growth through the creation of jobs and the spreading of economic activity (Guo and Jiang 2021; Dreher et al. 2022; Bluhm et al. 2025). Higher incomes in recipient-country populations lead to higher standards of living, which may result in more satisfied citizens. In turn, this may result in gratitude toward the donor government. Rather than using aggregate measures of income or remote-sensing measures such as nighttime lights, we measure income at the household level. Specifically, we analyze the logged self-reported per capita annual income in international dollars in column 1 and perceived income on a 4-point scale that ranges from 1 (“Finding it very difficult on present income”) to 4 (“Living comfortably on present income”) in column 2 of table 3. The results show that both actual and perceived income increase after the completion of Chinese development projects in recipient countries. An additional completed project increases the reported income by 0.4% and perceived income by 0.004 on the 4-point scale. Turning to

⁵⁹ Specific values are 22 (all projects), 28 (ODA projects), and 15 (large projects). The first-stage F -statistics remain strong when we compute Kleibergen-Paap F -statistics that are robust to heteroscedasticity, autocorrelation, and clustering (Olea and Pflueger 2013).

⁶⁰ Note, however, that the province-level results are not robust to how we define Chinese development funding. As can be seen in table D5, coefficients at the province level do not reach statistical significance when we replace the number of projects with a binary indicator for any project or measure it in (log) USD amounts.

TABLE 3
TESTING MECHANISMS, INSTRUMENTAL VARIABLE RESULTS

	Income (1)	Perceived Income (2)	Living Standard Up (3)	Living Standard Down (4)	Living Standard Good (5)	Community Basics (6)	Corruption (7)	Migration (8)	Environment (9)
A. Country Level									
Chinese projects _{t-1}	.00397* (.00236)	.00364** (.00147)	.000181 (.000604)	.000454 (.000786)	.00152** (.000584)	.000896** (.000415)	-.00118*** (.000433)	6.74×10^{-5} (.000399)	-.000435 (.000552)
Observations	417	481	472	472	484	462	470	474	473
Number of countries	90	95	94	94	95	93	94	94	94
Mean of dependent variable	7.419	2.455	.448	.249	.532	.562	.784	.599	.494
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
F-statistic	328.6	329.6	333.2	333.2	328.8	316.9	328.7	332.1	338

B. Province Level										
Chinese projects _{t-1}	.225 (.200)	-.210** (.0948)	-.0619 (.0395)	.0603* (.0344)	-.0408 (.0464)	.0354 (.0277)	.0231 (.0377)	.0511 (.0446)	.0203 (.0480)	
Observations	6,037	6,686	6,478	6,478	6,705	6,405	6,549	6,517	6,542	
Number of countries	91	95	94	94	95	93	94	94	94	
Number of provinces	1,415	1,465	1,439	1,439	1,465	1,444	1,443	1,456	1,448	
Mean of dependent variable	7.540	2.476	.433	.249	.543	.577	.799	.590	.495	
Country-year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Province FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
F-statistic	5.037	21.61	21.37	21.37	21.97	12.18	22.59	23.74	21.22	

Note. Shown are regression results at the country level following eq. (5) and at the province level following eq. (3), where we change the dependent variable as indicated in the column heads. "Income" is respondents' average logged self-reported per capita annual income in international dollars. "Perceived income" is respondents' average perception of household income on a scale from 1 ("Finding it very difficult on present income") to 4 ("Living comfortably on present income"). "Living standard up (down)" measures the share of individuals that indicate that their standard of living is going up (down). "Living standard good" measures the share of respondents satisfied with their current standard of living. "Community basics" is an index taken from the GWP for everyday life in the community, including education, housing, and infrastructure (we do not include the responses "Don't know" and "NA"). "Corruption" is the share of respondents that replied "yes" to the following question: "Is corruption widespread within businesses located in (this country) or not?" "Migration" is the share of respondents that replied "yes" to the following question: "Is the city or area where you live a good place or not a good place to live for immigrants from other countries?" "Environment" is the share of respondents that replied "yes" to the following question: "In the city or area where you live, are you satisfied or dissatisfied with efforts to preserve the environment." All specifications include the control variables age, age squared, gender, education, and urban in addition to the set of fixed effects indicated in the table. Standard errors (in parentheses) are clustered by country.

* $p < .10$.

** $p < .05$.

*** $p < .01$.

perceived living standards, we find no significant overall effects on the perceived change of living standards (cols. 3 and 4), but significantly more citizens report their living standard as good (col. 5). In the year after a project's completion, citizens are 0.15% more likely to express that they have a good living standard. Although these quantitative effects are modest, this first set of results supports the idea that better income and living conditions are a channel that links Chinese development projects and support for the Chinese government.

Second, development projects might improve the delivery of public services in recipient countries. For example, foreign-financed education and health projects might improve social welfare, education outcomes, or health conditions of the recipient population (Martorano, Metzger, and Sanfilippo 2020; Cruzatti, Dreher, and Matzat 2023). To test this, we use the Community Basics Index, which is the simple average of the share of respondents who express satisfaction with the following seven public amenities: education, health care, housing, water, air, roads, and public transport.⁶¹ As the results in column 6 of table 3 show, citizens are more likely to report satisfaction with community basics after the completion of Chinese projects. Public service provision thus appears to be a second channel that may link Chinese development projects and support for the Chinese government.

We also test three channels that might reduce approval rates of China's government. First, earlier research that focused on African countries finds evidence of increases in reported corruption around Chinese project sites (Brazys, Elkink, and Kelly 2017; Isaksson and Kotsadam 2018a). In stimulating economic activity and the availability of public and private goods, Chinese projects may increase demand for corruption. In turn, greater perceived corruption could lead to blame for China's government. On the other hand, improved living conditions might reduce individuals' willingness and opportunities to pay bribes.⁶² To test these possibilities, we analyze the effects of Chinese project events on the average share of individuals that confirm widespread corruption within businesses in their respective country. In line with Dreher et al. (2022), we find evidence of fewer rather than more reports of corruption after

⁶¹ More precisely, the index is computed on the basis of the answers to the following questions: "In the city or area where you live, are you satisfied or dissatisfied with [sector]?" ("educational system or the schools," "availability of quality health care," "availability of good affordable housing," "quality of water," "quality of air," "roads and highways," and "public transportation systems"). Only respondents answering all questions are included. We drop the answers "Don't know" and "NA" before calculating the index.

⁶² Such positive effects of income and development on corruption are well documented (e.g., Paldam 2021).

the completion of Chinese projects. The absence of corruption-enhancing effects of Chinese development projects may also help explain their overall positive effects on Chinese government approval.

Second, popular accounts often suggest that many Chinese development projects rely heavily on Chinese rather than local labor (Cervellati et al. 2022). While a large share of this labor is used during the implementation stage of the project, some staff, such as project managers and maintenance workers, may remain on the project after its completion. Potential resentment toward foreign labor in general, and reservations toward Chinese migrant workers in particular, might help explain negative sentiments toward the Chinese government. We test this by looking at citizens' attitudes toward migration (in col. 8). More precisely, we compute the share of respondents that confirm the city or area where they live is a good place for immigrants from other countries. We do not find evidence for a change in the attitudes toward migration. Again, the absence of anti-immigration sentiments may help explain the overall positive effects on Chinese government approval.

Finally, Chinese development activities might induce environmental degradation. Baehr, BenYishay, and Parks (2023) find that Chinese development projects lead to forest cover loss in Cambodia and Tanzania. Construction-heavy, large-scale infrastructure, mining, and energy projects are especially likely to cause damage to the natural environment in project regions through heavy use of soil and water, pollution, soil degradation, and deforestation. We measure satisfaction with environmental protection on the basis of the question "In the city or area where you live, are you satisfied or dissatisfied with efforts to preserve the environment?" (in col. 9). We do not observe significant changes in attitudes toward satisfaction with environmental protection in the aftermath of completed Chinese development projects. Environmental damage does not seem to harm the formation of more positive attitudes toward the Chinese government in recipient countries.

In panel B of table 3, we report the corresponding province-level results following equation (3) for the same dependent variables. Again, these results have to be interpreted relative to the overall country effects (which are absorbed by the country-year fixed effects). We find that individuals living in project provinces feel worse off, controlling for country-level effects. They report their living standard to be deteriorating and experience greater difficulties to live on their current income. In contrast to Isaksson and Kotsadam (2018a), however, we do not find evidence for higher reported corruption in businesses or the government in project provinces. Attitudes toward migration also do not change, and individuals living in project provinces do not report more or less satisfaction with communal or environmental services.

Deteriorating living standards relative to the country level might be one explanation for why the effect of Chinese development projects on popular support for the Chinese government is lower in provinces that host projects. However, the tests for mechanisms do not provide direct evidence for why this is the case. We suggest several possible explanations. One possibility is that project quality might be lower than originally expected, and this quality difference might be more strongly experienced locally in provinces with project sites.⁶³ Within provinces, individuals are able to compare their firsthand experiences and observations with their *ex ante* expectations, which might lead to negative *ex post* evaluations. The jobs, revenues, goods, and services generated by Chinese-government-financed development projects might not meet the expectations that were created when the projects were announced or initiated. Mismatches between *ex ante* expectations and *ex post* evaluations are perhaps less common, or at least less striking, among individuals not living near project sites and lacking direct, visible exposure to project outcomes.

Alternatively, perceptions in provinces receiving Chinese projects might also be less positive because of fungibility. If Chinese government funding supports projects that would have otherwise been supported by a different financier, and if the alternative project was expected to be of a higher quality or of greater importance to the people living in that area, individuals in project provinces might hold more negative views of the Chinese government in project areas compared with the country level (Cruzatti, Dreher, and Matzat 2023). The scale of our study and data availability make it difficult to pin down these channels. Future research could look specifically at identifying the mechanisms behind the differential province-level and country-level effects in the short and longer runs.

In short, though the effect on the support for China's government is lower in project provinces, we find overall positive, country-level public opinion gains for China's government. This latter finding is arguably more consequential for donors looking to use aid as a soft-power instrument. However, to comprehensively gauge whether aid promotes foreign support for donor governments, one must also consider whether development projects shape attitudes beyond the host country. The following section thus turns to third-country and global effects of Chinese development projects.

⁶³ Relatedly, negative spillovers—such as increased levels of corruption and environmental degradation—might be greater than originally anticipated. These points can also explain the difference in effects that we find between the short and longer runs.

V. Perceptions of the Chinese Government in the Global South

A net assessment of the potential soft-power effects of aid must also account for how development activities are perceived in countries other than the recipient country itself. As such, in addition to provincial-level and national-level tests discussed above, we also consider whether Chinese development projects shape attitudes toward the Chinese government at the global level. Development projects do not exist in a vacuum and can often be observed and assessed by individuals who live far away from where projects take place. These observations do not stop at national borders, and information about individual development projects often resonates with international audiences. Of course, in the process of doing so, this information can be repackaged, distorted, and presented differently to different audiences. Sri Lanka's notorious Hambantota Port, the "birthplace" of the "debt-trap diplomacy" narrative, offers perhaps the best-known illustration in the context of the BRI (Brautigam 2020).

International, third-country reactions to overseas development projects are important for donor governments. While donors hope to win hearts and minds within the national and subnational jurisdictions where they finance development projects, they also care about how such activities are viewed on a global scale. We therefore analyze whether our findings at the country level translate into global support for the Chinese government. If yes, then earlier research only considering local or national public opinion effects might underestimate the attitudinal gains of Chinese development finance for China's governments. If not, the positive recipient-country effects might instead be offset by negative global reactions to Chinese financing from a global soft-power perspective.

We test the public opinion effects of Chinese development projects at the global level by relying on the macro-level instrumental variables framework.⁶⁴ To this end, we run a modified version of equations (4) and (5) and estimate the effects of projects completed anywhere around the world (rather than only in the recipient country itself) on the country-level approval rate of the Chinese government. We aggregate the predicted number of projects in all provinces in a year to the global level. Because these aggregates are a time series without cross-sectional variation, we omit year fixed effects because such fixed effects would be perfectly correlated with the predicted number of projects worldwide.⁶⁵

⁶⁴ We focus on the longer term, as our micro-level analysis does not allow identification of the effects of (many) events that hit all countries globally because of the lack of an adequate control group. In addition, we believe that the longer term is more relevant for analyzing global soft power.

⁶⁵ Alternatively, one might consider to directly estimate 2SLS regressions at the global level. Given that input materials would then only vary over time—and fixed effects for years would thus have to be excluded—this approach arguably violates the exclusion restriction (many variables that vary from year to

Table 4 shows the results. As can be seen in column 1 of panel A, projects completed globally do not affect public approval of the Chinese government. The positive effects we find at the country level are either not sufficiently large to affect global public opinion about the Chinese government or positive effects are canceled out by negative opinions about development finance elsewhere. Either way, China's development finance does not significantly affect global approval of its government. The remaining columns of table 4 focus on subsets of countries that we consider to be of particular importance for soft-power acquisition to either China's government or its competitors for global influence: recipients of China's aid in the sample period (col. 2); frequent recipients of China's aid (defined as countries that receive aid in more years than the median value of three; col. 3); the world's least developed countries (col. 4); countries in Asia (col. 5); countries in Africa (col. 6); countries with a low voting alignment with China in the United Nations General Assembly (defined as the lowest tercile of voting similarity; col. 7), a middle voting alignment (col. 8), or a high voting alignment (col. 9); and countries with a low, middle, or high approval of the Chinese government (again defined as terciles, of the average approval rate over the sample period; cols. 10–12).

The results show positive and significant effects for the samples of African countries, politically neutral countries in terms of United Nations voting alignment, and countries with an already high level of approval of the Chinese government. By contrast, countries with an already low opinion of the Chinese government show a further reduction in their approval rate in response to Chinese global development finance activities. This suggests that opinions about the Chinese government become more polarized as China provides more projects. Quantitatively, 51 additional projects in the previous year improve support for the Chinese government on the African continent by 1 percentage point. On the basis of the median number of new projects per year worldwide (of 122), this estimate implies an aid-induced increase by more than 2.2 percentage points. Likewise, 56 additional projects in the previous year improve support for the Chinese government in "swing states"—those that are in the middle tercile according to their voting alignment with China in the United Nations—by 1 percentage point. The same holds for 54 projects in countries that have a high *ex ante* approval of the Chinese government. The negative effect in countries that disapprove of the Chinese government is slightly stronger in absolute terms, with 42 projects leading to an additional

year are likely to be correlated with input materials and approval of the Chinese government). For what it is worth, panel C of table D3 shows that when we do so, coefficients are positive and insignificant, in line with the average results we obtain in our main analysis on the basis of aggregated predicted projects.

percentage point of disapproval. It thus seems that China's development projects increase its popular approval in three important subsets of countries.⁶⁶

Our global analysis also considers whether projects given to a particular set of countries affect public opinion differentially. The other panels of table 4 investigate whether public perceptions of Chinese development projects implemented abroad depend on who receives the projects. More precisely, panels B, C, and D investigate whether projects affect public opinion more strongly when they are given to countries "closer" to one's own. In panel B, we therefore give larger weight to projects completed in countries with a greater similarity in the voting behavior in the United Nations General Assembly by using an index provided by Voeten, Strezhnev, and Bailey (2009).⁶⁷ Panel C examines geographically proximate countries, measured by the inverted logarithm of the population-weighted geographic distance between two countries (data from Mayer and Zignago 2011). Panel D focuses on ethnic similarity by using the inverse distance-adjusted ethno-linguistic fractionalization index on dissimilarity in ethno-racial characteristics (Kolo 2012). Again, we do this for the global sample in column 1 as well as all subsamples in columns 2–12.⁶⁸

The results show no significant effect of the completion of Chinese development projects on public opinion globally, independent of how we weight projects (col. 1). However, if China values soft power among a certain target group of countries, such as its (frequent) aid recipients, African recipients, or Asian recipients (cols. 2–3 and 5–6), use of development projects as a soft-power tool can pay off: aid creates support for the Chinese government in third countries that are politically aligned with the recipient country. Development projects also create a more favorable opinion in politically aligned third countries that are either politically neutral or China's friends (cols. 8–9). When looking at the effect of Chinese development projects in third countries that are either geographically or ethnically proximate to the recipient country, results are

⁶⁶ In Africa, China seeks to pursue natural resources and access to markets (Dreher et al. 2018); swing states in the United Nations General Assembly are those likely to be most accessible to exchange a specific vote for aid (Fuchs, Nunnenkamp, and Öhler 2015); and countries with high approval are more likely to be allies of China than those where approval rates are low.

⁶⁷ We use the *S*-score, which measures the voting similarity between two countries in a given session. It is calculated as $S_{ab} = 1(Y_{av}Y_{bv}/V)$, where $v = 1, \dots, V$ indexes votes, a and b refer to two countries, and Y refers to votes, taking on one of three alternatives: yea ($Y = 1$), abstain ($Y = 2$), and nay ($Y = 3$). An *S*-score of 1 means complete agreement and a score of -1 total disagreement between the two countries. Also see Bailey, Strezhnev, and Voeten (2017) for further details.

⁶⁸ These regressions include the control variables age, age squared, gender, education, and urban, as well as fixed effects for countries and years. The inclusion of year fixed effects affects the interpretation of our results, given that they capture any average effects the projects might have on public opinion globally. We also control for projects given to the country of interview.

TABLE 4
CHINESE PROJECTS AND GOVERNMENT SUPPORT AT THE GLOBAL LEVEL: INSTRUMENTAL VARIABLES RESULTS

	All (1)	Aid Recipient (2)	Frequent Recipient (3)	LDCs (4)	Asia (5)	Africa (6)	Political Foes (7)	Political Neutral (8)	Political Friends (9)	Opinion: Low (10)	Opinion: Neutral (11)	Opinion: High (12)
	A. Global South											
Chinese projects worldwide _{<i>t-1</i>}	7.71×10^{-5} (6.38×10^{-5})	6.23×10^{-5} (7.12×10^{-5})	.000115 (9.26×10^{-5})	.000160 (.000103)	-.000145 (.000166)	.000196** (7.76×10^{-5})	1.07×10^{-5} (.000136)	.000180* (.000102)	3.04×10^{-5} (.000106)	-.000236** (.000113)	.000199 (.000133)	.000186** (7.75×10^{-5})
Observations	452	388	264	170	119	160	66	177	200	136	154	162
	B. Third Country (Projects Weighted by Political Proximity to Recipient Country)											
Chinese projects abroad _{<i>t-1</i>}	.00122 (.000777)	.00200** (.000906)	.00324*** (.00110)	.000291 (.00155)	.00407** (.00156)	.00478** (.00185)	-.00317 (.00226)	.00239* (.00134)	.00438*** (.00133)	-.00149 (.00185)	.000119 (.00118)	.00125 (.000785)
Observations	443	382	257	169	113	160	64	177	200	126	154	161

	C. Third Country (Projects Weighted by Geographic Proximity to Recipient Country)											
Chinese projects abroad _{<i>t-1</i>}	.00962 (.0140)	.0104 (.0162)	-.00566 (.0234)	.00774 (.0264)	-.109** (.0491)	.0298 (.0198)	.0568** (.0221)	.0175 (.0210)	-.00376 (.0200)	-.0247 (.0302)	.0303 (.0292)	.00838 (.0169)
Observations	449	388	263	169	119	160	64	177	200	132	154	161

	D. Third Country (Projects Weighted by Ethnic Proximity to Recipient Country)											
Chinese projects abroad _{<i>t-1</i>}	7.57×10^{-5} (6.62×10^{-5})	.000101 (6.85×10^{-5})	9.80×10^{-5} (9.17×10^{-5})	5.27×10^{-5} (.000114)	-.00183*** (.000576)	.000131 (9.02×10^{-5})	-.000101 (.000322)	.000189* (.000102)	8.55×10^{-5} (9.60×10^{-5})	-.000478* (.000275)	-6.42×10^{-5} (.000140)	6.81×10^{-5} (.000101)
Observations	449	388	263	169	119	160	64	177	200	132	154	161

Note. Reported are the results of the global analysis. Panel A shows the results from eq. (5), where we aggregate the number of projects to a time series at the global level. Column 1 ("All") reports the results for all countries in the sample. Columns 2–12 restrict the sample to subsets of countries. "Aid recipient" includes countries that have received any Chinese aid in the sample period. "Frequent recipient" includes countries that received aid in more than 3 years (median). "LDCs" includes the least developed countries according to the World Bank's income categories. "Asia" and "Africa" include countries from the respective continent. "Political foes" ("Political friends") include those countries in the lowest (middle; highest) tercile of mean political agreement with China in the United Nations General Assembly (UNGA). "Opinion: low" ("Opinion: neutral"; "Opinion: high") is those countries in the lowest (middle; highest) tercile of mean approval of the Chinese government. Panels B–D report results for the number of projects completed in third countries (Chinese projects abroad_{*t-1*}). We aggregate the province-level predictions from eq. (3) to the global level, excluding the country of interview. Projects are weighted by using the following spatial weights: "Political proximity," "Geographic proximity," and "Ethnic proximity" weigh the number of projects completed abroad by similarity in terms of UNGA voting alignment, geographic distance, and ethnic similarity. All specifications include the control variables age, age squared, gender, education, and urban. Panel A includes country fixed effects. Panels B–D include country and year fixed effects and control for the number of projects completed in the country of interview. Kleibergen-Paap *F*-statistics are above 10 in all specifications. Standard errors (in parentheses) are clustered by country. LDCs = least developed countries.

* $p < .10$.
 ** $p < .05$.
 *** $p < .01$.

mixed. For example, while China can attract neutral countries through projects in ethnically aligned third countries (col. 8), such projects further reduce support in low-opinion countries if projects are carried out in Asia and ethnically proximate countries (cols. 5 and 10). Taken together, Chinese development finance leads to soft-power gains in important third countries when these are politically aligned with the recipient country. This provides the Chinese government with a tool to simultaneously enhance its influence among groups of aligned countries. In the next section, we examine whether Chinese development finance also boosts popular support for recipient governments.

VI. Is Aid “Win-Win Cooperation” for Donor and Host Governments?

In addition to attitudes toward donors, the completion of development projects may affect citizens’ views of their own government. Chinese development finance is often framed as “win-win cooperation” by the Chinese government, and if this principle holds true, we might also expect host governments to gain popularity for securing and implementing Chinese-financed projects. This contrasts with the conventional wisdom, which suggests that foreign aid may actually undermine state legitimacy. Many scholars have argued that when governments rely on foreign aid (or other sources of “unearned income”), they are insulated from the consequences of poor policy decisions and more easily able to ignore taxpayers’ preferences, effectively short-circuiting the accountability relationship between the governor and the governed (Djankov, Montalvo, and Reynal-Querol 2008; Smith 2008; Bueno de Mesquita and Smith 2010). However, a new wave of empirical research calls the conventional wisdom into question. Dietrich and Winters (2015), Dietrich, Mahmud, and Winters (2018), and Blair and Roessler (2021) provide experimental evidence that the local receipt of foreign aid actually improves trust in government and citizen perceptions of the state. Marx (2018) finds that incumbent governments benefit from the implementation of aid projects. Similar effects in the case of Chinese development finance would be in line with the Chinese government’s claims of “win-win cooperation” from the perspective of recipient-country governments.

Table 5 reports results of tests for whether Chinese development projects affect citizens’ support of their own government. We show results for both the micro and macro level, changing the dependent variable to the approval of the national government of the recipient country.⁶⁹ Column 1 reports results for our event study on the basis of equation (1). Column 2 is based on the same equation but includes a binary variable indicating the project province.

⁶⁹ Specifically, government approval is based on the question “Do you approve or disapprove the job performance of the (leader/head/president) of this country?”

TABLE 5
CHINESE PROJECTS AND APPROVAL OF THE NATIONAL GOVERNMENT

	Short-Term Country (1)	Short-Term Province (2)	Longer-Term Country (3)	Longer-Term Province (4)
Post/project completed	.0115 (.0178)	-.00148 (.0180)	.00210*** (.000791)	.0766* (.0432)
Post × project province		.116*** (.0437)		
Observations	18,994	18,994	420	5,920
Number of countries	19	19	86	86
Number of provinces	248	248		1,325
Number of projects	21	21		
Province-year FE	✓	✓		
Country FE	✓	✓	✓	
Year FE	✓	✓	✓	
Country-year FE	✓	✓		✓
Province FE	✓	✓		✓
F-statistic			341	20.26

Note. Reported are results for the effect of Chinese development projects on government approval in recipient countries. Columns 1 and 2 report short-term results from eq. (1). The dependent variable is binary and indicates whether or not an interviewed individual approves of the national government on the basis of the question "Do you approve or disapprove the job performance of the (leader/head/president) of this country?" The sample is restricted to individuals interviewed within 30 days before or after ("Post") a project-related event date. Columns 3 and 4 report the longer-term results on the basis of eq. (3) and eq. (5). The dependent variable is the share of individuals that approve of the national government in a given province/country on the basis of the same question. The variable of interest is the number of Chinese development projects completed in the previous year. Standard errors (in parentheses) are clustered by country-date in cols. 1 and 2 and by country in cols. 3 and 4.

* $p < .10$.

*** $p < .01$.

Columns 3 and 4 report the macro-level results at the country level on the basis of equation (5) and at the province level on the basis of equation (3), respectively.

Starting with the event-study results, we observe positive public opinion effects for the national government in project provinces (col. 2) but not elsewhere in the country (col. 1). Government approval rises by 11.6 percentage points after the completion of a Chinese project in the project province. Comparing these effects with those on Chinese government approval, it appears that short-term popularity gains for the national government are locally constrained, whereas support for the Chinese government improves countrywide. This is an important result for governments that seek to increase their support base in the short run (e.g., prior to elections).

Turning to longer-term public opinion effects, we present macro-level results in columns 3 and 4. Approval rates of one's own government improve with the number of Chinese projects completed, both at the province level and country level. An additional project increases the approval rate of the national government by 0.2 percentage points. While this effect is small

compared with the sample average of 54.5%, it is comparable in size with the effect on the approval of the Chinese government as reported in table 2. At the province level, the results are stronger with an additional Chinese project leading to a popularity gain of 7.7%. It indeed appears that Chinese development projects are “win-win” for both the Chinese government and the national governments of recipient countries, at least in the year after project completion.

VII. Conclusion

This study provides a comprehensive analysis of whether and how foreign aid shapes public support for donor governments in developing countries. To address this question, we created a new dataset of Chinese development project events (including precise project commitment, start, and end dates) and merged these data with granular, time-stamped data from the GWP in 126 countries and more than 2,000 subnational jurisdictions. We chose to focus on Chinese development finance and its effects on foreign popular support for Beijing for several reasons: China’s growing importance as an international donor and lender, its active pursuit of soft power, and the availability of high-quality, granular data on treatments and outcomes. Unlike previous analyses that measure only localized, short-run public opinion effects near project sites, our approach provides a comprehensive picture of the public opinion effects of Chinese development projects by considering different time horizons and levels of geographic aggregation (provinces, countries, and groups of countries). Our analysis is underpinned by two causal identification strategies: a short-term event study wherein the timing of interviews can be considered as random relative to Chinese project events, and an annual analysis using an established instrumental variables approach that exploits changes in China’s overall production of project inputs and the differential effect of these shocks on different countries and subnational jurisdictions.

Our results show that, on average, public approval of the Chinese government increases in countries where Chinese development projects are completed. In the short run, this effect increases with the size of the project and the generosity of the financial commitment; in the longer run, it is lower among people who live in close proximity to completed Chinese development projects. Perhaps most important, we find that these development projects create a more favorable public opinion environment for China among countries in Africa, potential “swing states” in the United Nations General Assembly, and countries with higher baseline (*ex ante*) levels of public support for the Chinese government. We also find that Chinese development finance leads to soft-power gains in important third countries—when these are politically aligned with the recipient country. These positive spillover effects allow the Chinese government

to simultaneously enhance its public profile among groups of politically aligned countries. In short, we find that China's overseas development program enhances its soft power.

At the same time, our findings help clarify a key source of confusion among policy makers, journalists, and scholars: whether the BRI is a reputational asset or liability. The existing debate about the BRI suggests that its public opinion effects are either overwhelmingly positive or negative. One camp argues that Beijing is gaining the upper hand in a zero-sum, great-power competition for international influence by bankrolling big-ticket infrastructure projects that Western powers are unwilling to support. Another camp claims that Beijing is losing the battle for "hearts and minds" because of local exposure to the negative unintended consequences of Chinese development projects. Our findings suggest that neither camp is entirely right—or wrong. Consistent with previous studies, we find that individuals who live near completed Chinese development projects form less favorable views of China (than individuals from the same country in the same year). However, unlike previous studies, we measure the public opinion effects of Chinese development projects also outside of the sub-national jurisdictions where such projects take place. Our findings indicate that indirect treatment exposure improves public sentiment toward China. These countervailing effects among individuals who experience direct treatment exposure and individuals who experience indirect treatment exposure appear to be a "net positive" at the country level, creating a more favorable public opinion environment for China.⁷⁰

The fact that those who are most directly affected by the costs and benefits of Chinese development projects develop less favorable views of the Chinese government, compared with the average effect in the same country and year, also helps explain why there are still many signs of BRI "backlash" around the globe.⁷¹ In this regard, our study complements previous studies, which provide evidence that those who live in close proximity to Chinese development projects have higher levels of exposure to various negative externalities—including political capture (Dreher et al. 2019a; Anaxagorou, Efthyvoulou, and Sarantides 2020), corruption (Isaksson and Kotsadam 2018a), ethnic tensions (Isaksson 2020), public protests (Iacoella et al. 2021), environmental degradation (Baehr, BenYishay, and Parks 2023), and declining rates of labor union involvement (Isaksson and Kotsadam 2018b).

⁷⁰ That being said, a limitation of our study is that it does not capture the public opinion effects of Chinese-government-financed development projects in the Global North.

⁷¹ Malik et al. (2021) find that at least 35% of the BRI infrastructure project portfolio has encountered major implementation problems, such as corruption scandals, labor law violations, and public protests.

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